

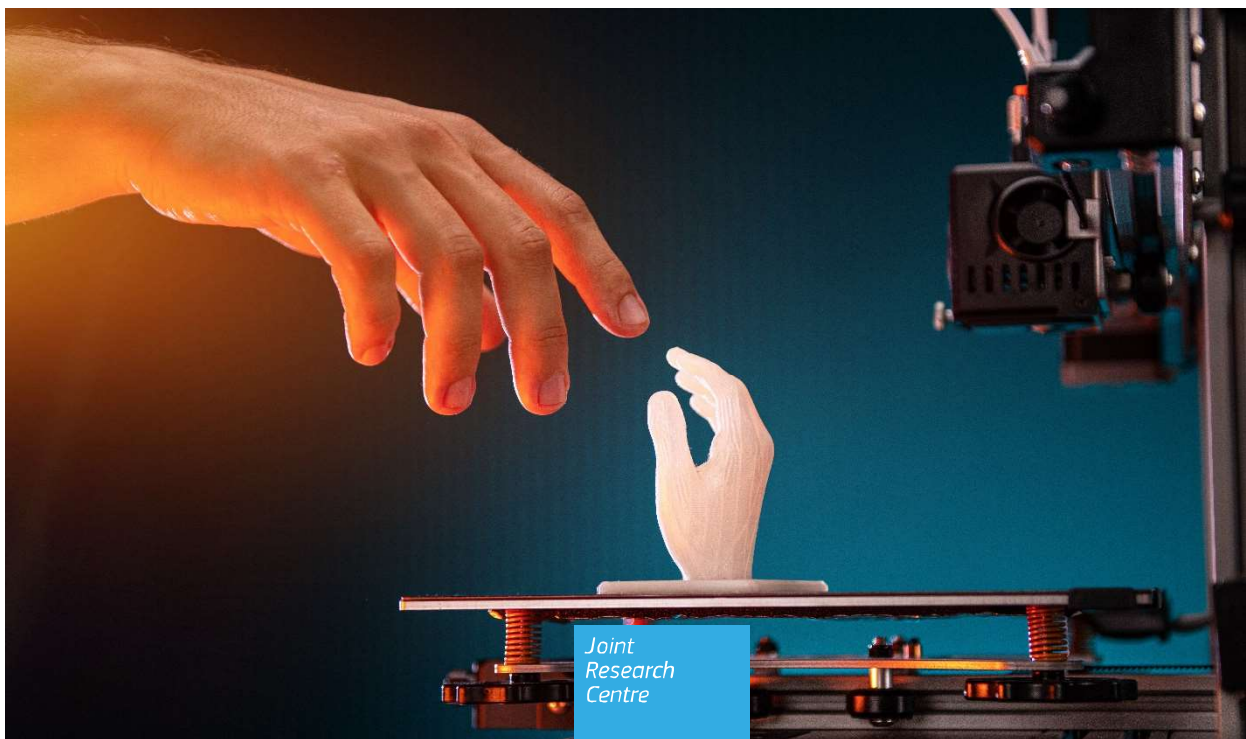
JRC SCIENCE FOR POLICY REPORT

Innovation Ecosystems in the Creative Sector: The Case of Additive Manufacturing and Advanced Materials for Design

Startups, their Emerging Business Models and their relationship with the CCI environment

Ubaldo Spina, Ramón Compañó

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Contents

1	Introduction.....	7
2	Novel Technologies for Design	8
2.1	Additive Manufacturing	8
2.1.1	Furniture, Lightning and Interiors	12
2.1.2	Textile-Clothing-Footwear, Jewellery, Fashion Accessories	13
2.2	Advanced Materials for Creative Industries.....	15
2.2.1	Advanced materials and its use in the creative industries	16
3	Emerging Trends affecting CCI	21
3.1	Sustainability	21
3.2	Product and Service Differentiation	23
3.2.1	Types of Classification and their Competitive Advantage	24
3.2.2	Custom versus Bespoke Design	25
3.3	CCIs as a revived opportunity for the development of rural areas.....	27
4	Analysing the Industry Landscape.....	30
4.1	Mapping Innovative CCI Companies	30
4.2	Mapping by CCI Sectors.....	31
4.3	Digital Creative Industries.....	31
4.3.1	New business models	33
4.3.2	Consumer as producers, as integrator of resources and as remixer	36
4.4	Emerging Trends	37
4.4.1	Survey results	37
4.4.2	Emerging Hubs	39
4.4.3	Original keywords	45
	List of abbreviations and definitions	48
	List of boxes.....	49

Abstract

This report provides an ecosystem analysis of cultural and creative industry (CCI) sector with regard to the adoption of new technologies and as case studies additive manufacturing and advanced materials as these two technologies are becoming consolidated in prototyping and manufacturing in many industrial sectors, but are still in their infancy in the CCI. The introduction of these two technologies, however, is the base of emerging new business models as showcased by a number of promising startups. We identified 430 CCI-startups in Europe that operate innovative products or services and that were further analysed via an online survey to which 83 companies participated. In addition, we discuss the role innovation ecosystems for the CCI development in Europe taking four ecosystems as example, namely the 'Finnish Additive Manufacturing Ecosystem' in Helsinki, the 'Design Farm' in Berlin, the 'Sustainable and Tech Fashion' hub in Amsterdam and "Portuguese footwear cluster" in the northern region of Portugal. Our study shows that green and digital transformations make additive manufacturing and advanced materials technologies quite worth looking into for CCI firms. The survey demonstrates a high degree of business dynamism of those companies that have fully embraced additive manufacturing and advanced materials as their operational basis. Furthermore we show that an effective ecosystem are key for deploying novel technologies to the sector, mostly composed by small and very small enterprises.

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Authors

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Executive summary

Policy context. Cultural and Creative Industries (CCI) are an essential part for the continued development of societies and important part of Europe's economy. CCIs are knowledge-intensive and largely based on individual creativity and talent, generating growth and create jobs. But CCIs do not escape from the techno-societal or circular economy megatrend: they have to master the green and digital transition, too. In this respect, a number of EU cities and regions did already start the transition and have been very successful at tapping into the potential of CCIs to boost their socio-economic development. However, there are still too many places in Europe, which have not yet been making most of their potential.

Many industrial sectors and value chains will have to go through a very rapid green and digital transformation in the next years towards a sustainable model of inclusive growth, embracing an improved environmental footprint along all steps of the value chain. Within this industrial context, investments in strategic value chains and new forms of collaboration with industry will be essential for ensuring Europe's strategic autonomy and technological leadership. Likewise, in order to support the implementation of European value chains across EU territories, it is necessary to understand the innovation capacity in particular places, the dynamics of their actors and whether & how effective working relationships can be established across Europe.

Technological Considerations. Additive manufacturing (AddManu) is increasingly finding its place in the creative industries as it allows them to move from custom production towards mass-customisation, to design products with previously unattainable geometry, to reduce the manufacturing complexity, to improve material efficiency and to simplify the supply-chain. We present two dozens of startups that show how AddManu is finding its way for interiors, particularly furniture and lightning and for fashion, the application areas are clothing, footwear, and jewellery. In a survey we carried out we find that AddManu is mostly employed in design stage, and that conceptual design is the process step expected to benefit most in the mid-term.

In this study we analyse also CCI startups that employ advanced materials (AdvMat) in their manufacturing process, such as advanced composites, textiles & fibres, coatings, gels & foams, high-performance polymers, light alloys or nanomaterials. We exemplify our results using a dozen of startups for which advanced material constitute the backbone of their businesses. For many years the introduction of advanced materials was on grounds of technical performance and costs. More recently, social and environmental motivations prevail.

Trends. Responsible consumption patterns and sustainability considerations come increasingly into the fore. In fashion, clients are increasingly demanding sustainable "cradle to grave" solutions, assuring the recovery and recycling dynamics set from the start of the design process. Such concepts can go beyond mere recycling, such as using fabrics that can be easily be dissolved, transformed or re-polymerised into new semi-finished products.

Another aspect influencing the choice of materials is the enhanced sensitivity towards animal protection and more people practising veganism, calling for new natural fibres and materials. Examples include (i) vegan textiles made of mushroom skin and mushroom roots, (ii) naturally dyed sports shoes grown with microbes, which lose more than 70% of their mass one week of after composting or (iii) wool fibre created through biodegradable fibres.

This trend may become of interest for rural areas. As ‘providers of novel materials’ for CCI industries may appear to collect, select, shred and mix significant quantities of post-consumer materials or agro-industry waste, developing compounds suitable for specific applications. CCIs in rural areas may then move up the value chain becoming ‘digital artisans’ managing the whole design service, i.e. analysing client’s needs to process fashion accessories or goods for home living.

Emerging Business Models. Introducing additive manufacturing tools can change the company’s former design and manufacturing process. Likewise, materials with additional functionalities or sustainable materials serve to offer new kind of products. From a business point of view a number of conclusions can be drawn. First, new services and products can lead to new business opportunities for the CCIs.

Second, AddManu and AdvMat in combination with the CCI’s digitalization will change the interaction between the stakeholders. For instance, online customer platform can change the speed and the degree of interaction between the former CCI’s providers and final consumers (prosumer). While the overall process remains as before, digitalization changes the roles and activities of the stakeholders, which become more interlinked. In a traditional business a client would enter a shop to buy a product; let say a designer table. The CCI would design the table, manufacture it and sell it. In the new setting the client can take active part in the design of the product (e.g. detailing some characteristics of the table) and adapt it to the customer’s specific needs (e.g. the dimension of the table). The client becomes a ‘prosumer’.

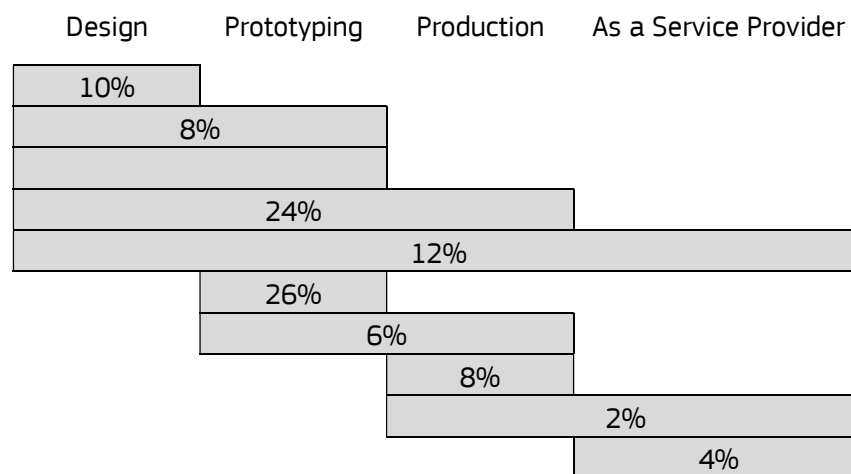
From a business model point of view, the novelty lies in the separation of design, delivery, adaptation and production as independent building blocks, which allow for potentially generating revenue streams from each of the building block. The figure underneath describes the stakeholder’s actions, the actions, the required infrastructure and the revenue model for each step.

	Design	Delivery	Adaptation	Production
Stakeholder	Designer	Marketplace	Prosumer	Local Manufacturer
Stakeholder Structure	Autonomous or SME	IT Company	Individual	SME
Infrastructure	Design tools	Web Platform	Personal Computer	Manufacturing tools (e.g. CNC, 3D printers, etc.)
Actions	IP-Creation To create concepts, projects, models, etc. that bear a IP	Dissemination To generate Visibility, promotes products	Adaptation To adapt the designer’s IP specifically to the consumer needs (e.g. the dimension of the table)	Manufacturing Manufacture Product upon the instruction of the Prosumer
Revenue Model	Licencing (e.g. royalty based)	Transaction based (e.g. fixed or %)	N/A	Service-based, direct payment

Examining tech-based CCI Startups. We screened the CCI landscape in the Europe and identified 430 companies that design and/or manufacture innovative products using additive manufacturing techniques or advanced materials as described before. These companies are active in Fashion, Fashion Design, Furniture, Furniture Design, Home Living, Interior Design, Material Provider, Service Design, Technology Provider and Marketplace.

56,7% of the 83 companies that took part in the survey, run very personalized services,¹ mostly focusing on ‘bespoke design’,² followed by ‘custom design’ approaches, which shows that personalization services can be the key to economically successful CCI businesses. From an IP point of view, most companies operate as Original Design Manufacturer (43%), while only few exclusively as Original Brand Manufacturer (8.4%) or Original Equipment Manufacturer (8.4%).³

With regard to the use additive manufacturing tools, some companies use AddMan exclusively for one process step: design (8%), prototyping (26%), production (8%) or as a service provider (4%). The appearance of the role ‘service provider’ is a sign of the emergence of business model based on AddManu specialisation. Another finding is that the majority of companies employ AddManu for two, three or all four steps, indicating that this technology is becoming established expanding beyond its original realm of prototyping.



In this sample, 27 respondents use AddManu for producing finished products, of which 4 of them do it exclusively with AddManu. These are company’s development and direct sale of goods made in 3D Printing. In terms of the most widely used additive technologies, there is an equitable distribution of all those surveyed, with a prevalence for Fused Deposition Modelling technologies.

With regard to the degree of ‘internal’ digitalization, 39 out of 83 companies use advanced digital technologies in their production and services. And regarding ‘external’ Digital Customer Engagement, 26 subjects declare to involve customers within the production process, with the possibility of customization, choice or booking.

Furthermore, we investigated how digitalization is influencing CCIs. Amongst the ‘digital’ CCIs, we observe three kinds of startups. The first one are those, for which IT is the centrepiece of their business; they act as ‘digital technology agents’ for other creative industries. Examples include platform (marketplaces) or technology providers (artificial intelligence, internet of things, virtual reality, bionic systems, etc.). The second group are those CCIs have integrated digital technologies in their design and internal manufacturing processes. The third group are those that, additional to the internal processes, use IT for external ones and actively involve the customers in the design or production process.

¹ 43.3% of respondents are engaged in business to business (B2B) services, 21.7% in business to client (B2C) services and the remaining 35% operate both B2B and B2C.

² Custom design starts with a pre-existing template or model that can be adapted to the needs of an individual only within pre-defined limits. Contrary to custom design, in bespoke design products are "made-to-measure" or "made-to-order".

³ The remaining percentage are companies to which the question does not apply

Ecosystems. Innovative CCI startups seldom appear in isolation, rather they emerge as a spin-off of previously existing industries or in environments with a high density of creative talents. Several countries have realized the importance of nurturing the CCI ecosystem. Examples of geographical hubs that pursue contribute to strengthening an innovation driven ecosystem include the DesignFarmBerlin in Berlin (focusing on smart textile wearable technologies), the Finnish Additive Manufacturing Ecosystem in Vaasa (3D printed furniture and biocomposites for 3D printing), FAMEST in Madeira (sustainability and circular economy applied to biomechanics and sensed foodware), Porto Global Hub (a design factory for applied research and industrial collaboration), ClickNL in Eindhoven (diverse activities for local creative industries) or 'REG X' in Vejle in Southern Denmark, (providing IT services and design support for user experience and user interaction).

Policy Considerations. CCI ecosystems contribute to economic, societal and environmental policy challenges. It is claimed that they may contribute fighting against the depopulation of rural areas, avoiding migration to urban areas. We observe, however, that most innovative CCI startups are located in traditional innovation hotspots like Berlin or Amsterdam. It is in these metropolitan areas where an ecosystem exist where talented people and technology wise can meet and find the infrastructure, capacities and funds to advance promising projects. In rural areas where the density of talented people is lower, the importance of organizations promoting measures to tighten networks and support innovation are more critical. Examples of organization whose aim is to integrate specific local knowledge with the opportunities deriving from new technologies include FAMEST in Madeira, FAME in Vaasa or REG X in Vejle. Whether in metropolitan or rural areas, there seems to be need to support startups with hands-on practical training on technological, and particular digital, support for design, modelling and manufacturing.

Key conclusion. As whole, CCI industries still have a long way to go to master the green and digital transition. This study, however, shows that there are hundreds of innovative firms and startups in Europe that have already mastered this transition. They make use of state-of-the art technology (digital tools, additive manufacturing, use of advanced materials, etc.) in their respective business process steps that allow them to master societal and environmental demands (sustainable production, animal-free materials, recycled materials, etc.), while staying economically robust. The support of a functioning local, placed-based innovation ecosystem is key to promote and help companies which are still in the transition phase. The case studies presented in this study show that such ecosystems can emerge in different geographical environments taking into account the specificities and needs of the regions.

1 Introduction

Many industrial sectors and value chains will have to undergo a very rapid green and digital transformation in the coming years towards a sustainable model of inclusive growth, embracing an improved environmental footprint along all steps of the value chains. Within this industrial context, investments in strategic value chains and new forms of collaboration with industry will be essential for ensuring Europe's strategic autonomy and technological leadership. Likewise, in order to support the implementation of European value chains across EU territories, it is necessary to understand the innovation capacity in particular places, the dynamics of their actors and whether & how variable geometry coalitions across Europe can be established. Cultural and Creative Industries (CCI) are an essential part for the continued development of societies and important part of Europe's economy. CCIs are knowledge-intensive and largely based on individual creativity and talent, generating growth and create jobs. But neither CCIs do escape from the techno-societal or circular economy megatrend: they too have to master the green and digital transition. In this respect, a number of EU cities and regions did already start the transition and have been very successful at tapping into the potential of CCIs to boost their socio-economic development. However, there are still too many places in Europe, which have not yet been making most of their potential.

The aim of this report is to analyse a number of place-based innovation ecosystems in promising segments of the Creative Industry. The report reviews specifically the impact of additive manufacturing (AddManu) and advanced materials (AdvMat) on the design sector, including trends, novel business models, and interactions between stakeholders. We start by mapping innovative CCI startups and analyse the industry dynamics. Taking the industrial landscape into account we select meaningful case studies to shed light into the dynamics from an innovation ecosystem point of view. Our study aims at providing new insights in the functioning of existing ecosystems in CCI and presents good practices in terms of technology integration, network dynamics, governance, or internationalization.

2 Novel Technologies for Design

This section introduces additive manufacturing (AddManu) and on advanced materials (AdvMat) in the creative industries and specific aspects such as the classification of technologies / materials and a scouting of recent specific applications.

It describes the productive dynamics and the impact of technologies on manufacturing industries, identifying possible interactions with services with the objective to understand, to anticipate and to address the impact of new technologies on design and production processes and hereby to contribute to smooth transformation the role of designers and creatives in general. The three components of the study are:

1. The level of knowledge and maturity and scope of applicability of AddManu technologies and advanced materials, in terms of specific subsectors in Europe;
2. The (potential) qualitative impact on the production process, including the impact on value chains, business models, productivity and products / outputs;
3. the (potential) creative impact on projects, in terms of complexity, personalization, sustainability, types and levels of skills as well as physical places of production.

The first part introduces definitions, concepts, trajectories and evolution of the role of AddManu in the creative industries, motivating how the increasing automation is gradually shifting the focus from the process to the product, with an important empowerment of the creative professions (both in terms of planning and management of extended and multidisciplinary teams). The current trends in the Furniture, Lighting and Interiors sectors (i.e. the introduction of AddManu in the products that populate homes and improve daily life) and Textile-Clothing-Footwear, Jewellery, Fashion Accessories (or the introduction of AddManu in wearable products, generically attributable to the fashion sector).

The second part introduces scientific and functional definitions and trajectories of some classes of materials, classifiable precisely as advanced, with references to present and prospective applications in the sectors presided over by the creative industries concepts, trajectories and evolution of the role of AddManu in the creative industries. The time horizon is five years, that is, it collects publications and productions presented starting from 2017 and with potential previews until 2022.

2.1 Additive Manufacturing

Additive manufacturing (AddManu) processes are considered in many applications as a new industrial revolution. Technologies and processes currently used in CCI are manifold and diverse; and they are used in different combinations rendering a complex set of new tools for CCI. Over recent years, different terminologies have risen to encompass these technologies, such as “rapid prototyping”, “rapid tooling”, “3D printing”, and “freeform fabrication”. While all of these are commonly accepted, “additive manufacturing” is probably the term that captures best its essence and is the term that we will use in this study.

Box: Additive Manufacturing

In additive manufacturing (3D-printing), a three-dimensional object is constructed from a CAD model or a digital 3D model. A variety of processes can be used to deposit, to join or solidify material under computer control to create a three-dimensional object. The materials are typically added layer-by-layer and different types of materials can be used, including plastics, liquids or powder grains being, which are fused together

Additive manufacturing changes the way products of services are made and delivered and what this means for creative sectors. The introduction of AddManu in creative industries will impact value chains, business models and specific activities such as product design and prototyping. These changes require people ready to work with these technologies and that are open to changes in production processes. AddManu in the creative industries, although not new at all, is still considered an emerging technology because it is still not widely diffused and has not yet reached a maturity level across the sector.

While not yet consolidated, we observe a fast adoption of AddManu technologies is due to a miniaturization of machines, to an ever greater computing capacity, to an important expansion of the range of materials used and to the reduction of cost. In some cases this acceleration is propelled by the expiration of the first registered patents. In essence, AddManu is a new way of turning data into physical things, which enables creative industries to.

- To move from custom production towards mass-customisation;
- To design products with previously unattainable geometry;
- To reduce the manufacturing complexity;
- To gain material efficiency; and
- To simplify the supply-chain.

The main advantages of these technologies are the high precision, the possibility of using different materials (available for metals, ceramics, plastics and biomaterials), and the ability to obtain impossible prototypes using conventional means. AddManu is applied mostly in creative industries where the following parameters influence decisions about technologies and production processes:

- High added value of customisation;
- Small batch size;
- Design complexity;
- Importance of low weight;
- Drastic assembly reduction.

Therefore, AddManu is increasingly being applied in the creative industries. In addition, there are broader, 'horizontal' applications of AddManu to produce components for prototypes and spare parts. Especially for these applications, the technological developments and the cost effectiveness and price of AddManu (mostly for metal) are important drivers. Other drivers (or enablers) are the extent to which AddManu can deliver on parameters such as reducing weight and simplifying assembly lines. Current barriers include technological immaturity (depending on the material used), standardisation and interoperability issues, a lack of

certification,⁴ and to some extent IP concerns. Other limitations in design processes include the high cost of the processes, the time required to obtain the prototypes, and potentially a lower degree of resistance. Active work is under way to improve these limitations in order for additive manufacturing to be competitive with regard to other more conventional means.

According to a recent an online survey of product and industrial design practitioners,⁵ the typical size of components produced in series in AddManu varies from 10 x 10 x 7 mm to 400 x 500 x 280 mm. The average size of 138.7 x 111.8 x 61.2 mm is suitable for fashion, but too small home living applications. For end-user components, the typical volume to produce is in the order of 11 to 1.000 pieces. Generally speaking, small scale production suffer from high fixed costs and weak economies of scale. Cost reduction is particular essential in such cases and AddManu contributes to his objective. The extent to which AddManu reduces cost with traditional manufacturing depends of the product and the sector. In all cases, however, AddManu offers an unprecented alternative that suits both niche production as well as high-end markets with low volume.

Polyamide and ABS seems to be the materials frequently most used. The main reasons for using AddManu were low volume, complex shape, speed, cost, shape manufacturability and customisation. The main limitations of using AddManu for series production were cost, post processing, productivity, materials, build platform and accuracy. Cost is the main perceived reason for not employing AddManu in production. Interestingly, many designers have never considered AddManu-based design nor AddManu production.

The more the AddManu process becomes further user-friendly and require less expert knowledge for its handling, the more AddManu enables to shift towards design and creativity.

To which extent creative industries exploit DfAM methods (Design for Additive Manufacturing) to generate creative concepts, solely, or whether is also employed in developing new projects is an interesting aspect, which can be assessed by its impact on the product definition at different levels:⁶

Level 1: Formal newness – Here, DFAM methods are oriented to the redesign of existing products. Designers start the process knowing most of the product's data: they know their inner and outer functions,⁷ how are its inner and outer forms and the product behaviour. The purpose is to redesign in order to render the product suitable and optimized for AddManu.

Level 2: Functional reconfiguration – Methods of this category are dedicated to redesign existing products that embody assemblies, for instance producing several components in one go,⁸ or redesign multiple components into one. The purpose is to reduce the number of components of existing assemblies or of existing whole products.

⁴ For instance, broken or damaged components could be reproduced by AddMan, but not being produced by an original equipment manufacturer (OEM) the component is not certified and employing them would remove the legal liability of the manufacturer

⁵ Moultrie, J., Pradel, P., Zhu, Z., & Bibb, R. (2017). "Designing end use components for additive manufacturing - Exploring experiences, knowledge and information requirements". (ISBN: 978-0-903428-42-3) <https://doi.org/10.17863/CAM.20973>

⁶ Rias, C., Bouchard, F., Segonds and S. Abed, (2016). "design for additive manufacturing: a creative approach". INTERNATIONAL DESIGN CONFERENCE - DESIGN 2016 Dubrovnik - Croatia, May 16 - 19, 2016.

⁷ Inner functions are those that do not involve direct contact with the user, while outer functions are those that activate the product through user interfaces.

⁸ For instance AM allows to create hollow parts and overcome undercuts otherwise not achievable with traditional moulding techniques. Or joining several components into one piece by growing layers.

Level 3: **AddManu Form & Function implementation** – Methods of this category are intended to the design or redesign of products. The purpose of the methods is to globally emphasize the use of AddManu capabilities in product design.

In its survey, Moultrie et al. (2017) analyses the most suitable stages of the design process for DfAM and finds that conceptual design, embodiment design and detail design is the most prominent one with 21.8% of the answers. 18.2% at all four stages of the design process (including brief setting), while 17.3% only at the conceptual design stage. **Although DfAM is currently mainly employed at the detail design stage, the conceptual design is the process step expected to benefit most from introducing DfAM.** This result suggests to introduce AddManu knowledge as early as possible into the design process and particularly during idea generation. At the same time, time it has to be recalled that often AddManu is still not mature enough from the final stage near production and testing.

In any case, the role of the designer, understood as the creative person, gets reinforced. More time and efforts will be spent exploiting the potential of AddManu and directing the results towards new unexplored scenarios while reducing direct human involvement in the production process (many consider AddManu to be a full-fledged robotic process). As 3d printers are flexible tools to creatively deposit specific materials, it empowers people with printer control skills to develop creative businesses or even general purpose 'creative factories' beyond professional designers

These changes in the production process imply that, increasingly, the tasks and occupations of designers, material suppliers, IT specialists will be intertwined and that an efficient and dynamic collaboration promoted by multidisciplinary teams will therefore be necessary (examples in chapter 2 demonstrate that this enables new business models such as 'bridge manufacturing'). The more the "production" process gradually becomes safe and controllable, the more there will be a transition from process responsibility to product responsibility. This is an indicator of how the creative process is crucial and how the creatives themselves: people skilled in traditional abilities (sketch, drawing, project culture, etc.) -but 'IT illiterate'- are called upon to digitize their skills to make the most of the potential of AddManu.

In terms of the process, further considerations need to be made. Assuming that production activities do not require direct operator intervention and that AddManu tech allows for contributes to manufacture more complex projects in a single run (see the potential offered by multi-material printing), the challenge remains that part of the success of creative products comes from the surface finishing of consumer products, which cannot yet be automatised. In this sense, it is worth monitoring how additive manufacturing will evolve and how the work teams will handle the manual post-processing of the components, using traditional processes or contributing to the introduction of new ones.

In the following chapters, we present aspects connected to the methods and places of supply of services for the creative industries. At the moment, the relative complexity of domestic printing and the big dimensions of some products, such as furnishings, do not suggest an immediate future for the "click to materialize". Local production centres and producers' spaces, where local specialists and inventors meet, are a more realistic conception of the 'democratization' of production and of the transformation of creative industries. And this could be a revolution within the revolution.

Needless to remind that additive manufacturing is not the best technology for any purposes, however it is rather about what designers consider is the most suitable technologies for a specific use. For example, in the following sectors:

2.1.1 Furniture, Lightning and Interiors

By now, AddManu technologies are widespread and tested in the creative industries. However, whether its use is consolidated in the design and prototyping activities, is unexplored. Similarly, whether the transformation of some professional firms or SMEs occur rather in digital factories or in small centres of production of consumer goods is equally of great interest. Designers and manufacturers use 3D printing technologies to increase complexity, map out unique or unexplored visual identities, and offer customizable furniture. Within the following paragraph some experiences have been collected, providing information relating to furniture, lighting and interior design products, technologies and 3D printers, with some references to the hybridization between design / sustainability / approaches to market.

Box 1 - Startups using Additive Manufacturing technologies for furniture, lighting or interiors

Drawn Furniture - Drawn is a French company that has developed its XXL 3D printer called Galatée to create custom furniture and decoration. After a successful Kickstarter campaign in 2015, it started manufacturing plastic stools, tables and chairs and smaller decorative items such as vases. The users can choose the colour and shape of their furniture and watch the 3D printing of a chair, for example.

Nagami - The Spanish design studio Nagami presented a collection of avant-garde futuristic chairs printed in 3D from different materials; two of them made with biodegradable plastic granules. They remind us of a floral structure and are very colourful. This project demonstrates the possibilities offered by 3D technologies in the architecture and furniture sector.

Morgan Furniture + Studio Integrate - Integrate and Morgan worked together to create RIO, a collection of chairs and tables that integrate 3D printed elements. For example, it is the backrest of the chair that has been printed in 3D in 24 hours from a liquid resin or polyamide powder following a mathematical algorithm. The result is a unique piece of furniture that combines craftsmanship and new technologies. The same principle applies to tables where part of the base has been printed in 3D.

XtreeE - The French startup XtreeE is a concrete 3D printing specialist that develops advanced large-scale 3D printing technology for architectural design, engineering and the construction sector. XtreeE ran the 'woven concrete benches' project in which they created a series of three benches by collaborating with Berlin-based design studio, Studio 7.5. The benches exploited the woven pattern both for aesthetic value and its structural performances. It is a complex motive to obtain with a classic mould. The 3D printed design is both geometrically complex and uses a minimal amount of concrete.

Beatrice Müller - The BigRep ONE v3 was created to make industrial 3D printing for large-scale objects as easy as possible. The printer offers 1m³ print volume. 3D designer Beatrice Müller designed furniture to be printed almost in one piece on the machine. In BigRep's Ocke Series, the set of 3D printed furniture consist of a chair and sofa.

Zortrax – Zortrax, a 3D printing manufacturer, created the KARO coffee table with its M200 3D printer to showcase the potential of their technology. The coffee table's legs were 3D printed in the company's Z-ULTRAT black, grey and ivory filaments. The legs of the table add up to 3 kg of plastic. It took nearly 263 hours to print the several diamond-shaped and triangular modules needed to assemble the table. Once printed, a circular glass was placed over the base to form a classy tabletop. KARO is a weather-proof furniture piece, suitable for both indoor and outdoor home environments.

ALEKSA - London-based ALEKSA studio designed a table with additive legs manufactured by 3D Print using an SLS printer. The structure of the legs makes it possible to carry the heavy walnut wood top despite their minimal weight. The use of 3D printing technologies allows designers and manufacturers to produce different variations and sizes of the table on request.

Jon Christie - Jon Christie merges additive manufacturing technologies with traditional furniture manufacturing techniques. This is how his first collection emerged, taking a Scandinavian design of chairs has decided to add 3D printed gaskets, which has allowed greater creativity in terms of colour and shapes when creating. For its development it has used the melted deposition technology of the Dutch brand Ultimaker.

Christophe Guberan - The collaboration between MIT's Self-Assembly Lab, designer Christophe Guberan, and furniture brand Steelcase yielded to the development of a new 3D printing method: Rapid Liquid Printing. It draws effectively inside a gel the structures to be printed. It was thanks to this development that an original table was created, which takes advantage of the fact that this manufacturing technique has no gravity limitations and is capable of doing so in less time than traditional technologies. The table was printed in 28 minutes.

LPJacques - The French design studio LPJacques specialises in the use of additive manufacturing technologies for their projects. Di-Lamp is a project in which the lamps can be completely customised by users thanks to additive manufacturing technologies. People can choose the type of design and colour or in addition a series of decorative elements to match the lamp.

2.1.2 Textile-Clothing-Footwear, Jewellery, Fashion Accessories

In recent years, 3D printing technologies have penetrated the fashion industry, particularly the high fashion industry, to offer more innovation in designs. Given the design freedom afforded by additive manufacturing, it makes sense that designers are interested in working with this technology to create complex shapes and original pieces. From shoes to dresses, 3D printed fashion is very different from conventionally manufactured clothing. Each design tends to be unique, personalized and sustainable. This is another reason why 3D printing is prospectively very interesting for this sector.

3D printing opens up a wide range of opportunities for stylists to create innovative and futuristic designs. As in demanding industry applications, going beyond traditional design boundaries is no exception in the fashion world. According to researchers from the International Journal of Fashion Design, five types of 3D methods show real potential in

fashion: stereo-lithography, selective laser sintering, fused deposition modelling (FDM), PolyJet, and binder jetting. However, FDM and SLS seem to be the most used technology by designers. It should also be highlighted that in the creative industries, and in particular in the Textile-Clothing-Footwear sector, reducing waste is a particularly significant challenge for approaching increasingly responsible users and consumers and AddManu, in this sense, offers concrete answers for the creation of low environmental impact collections.

Finally, it should be noted that the production of footwear or specific parts or customizable insoles for everyday use, sports or specialized activities is one of the fastest growing segments of 3D printing.

Box 2 - Startups using Additive Manufacturing technologies for textiles, clothing, footwear, jewellery, fashion or accessories

Annie Foo - High-end Shoes Annie Foo is a designer of high-end shoes who uses 3D printing technologies in their manufacturing process. She combines existing footwear manufacturing methods with modern technologies and materials to create something new. Using 3D modelling software, she designed complex shapes to give her shoes an aesthetic look. Her final pieces, produced for catwalks, were made with HP's Jet Fusion 4200 machine, using PA12 (nylon).

The Proximity Dress - Anouk Wipprecht is a Dutch designer known for combining new technology and fashion to create unique clothing with unexpected features. This is the case of her new dress called Proximity Dress. Anouk imagined it in the context of the COVID-19 health crisis: the dress includes different sensors that are able to detect movement next to the wearer. If a person gets too close to the wearer of the dress, the dress expands to create a kind of barrier. A clever idea developed in part through 3D printing: Anouk used the SLS process as well as Stratasys' PolyJet technology to manufacture various components of the Proximity Dress.

Julia Körner - The designer and founder of JK Design GmbH developed the first 3D printed dress together with Iris Van Herpen and the company Materialise. Her latest collection is called "Chro-Morpho", which was created in collaboration with Stratasys and is intended to demonstrate the opportunities and diversity of 3D printing in the fashion world. Nature is a great source of inspiration in the SETAE jacket.

VIP TIE - Is an Italian company that is famous for 3D printing ties. The goal of the company is to offer fashionable accessories that are as personalized as possible in order to satisfy the needs and desires of the customers. VIP TIE combines both luxury and innovative high-tech design. The creation process of just one tie can include an intricate mix of several complementary procedures and materials: craftsmanship, embroidery, 3D printing, exotic leather, mother-of-pearl, carbon fibre, silver, gold and silver plated solid. Apart from highly customizable products, another advantage of VIP TIE is that more than 80% of the materials used are 100% environmentally friendly.

VOJD Studios - Berlin-based startup VOJD Studios creates 3D printed accessories and jewelry to challenge the perceptions of conformity. The company's vision is to promote a synergy between luxury and progression. Therefore, the label combines the latest technological innovations with traditional quality craftsmanship to redefine the way people think about design. In addition to its jewelry collections, VOJD also works with luxury fashion labels such as Kenzo, Acne Studios, Louis Vuitton, Alexander McQueen, and Carolina Herrera and helps them integrate 3D printing in their collections. As for technology, VOJD mainly works with SLS, DMLS, binder-jetting and SLA technologies and uses a wide range of materials such as silver, steel, ceramics, nylon, resin, etc.

XYZBAG - Based in Turin, XYZBAG is an Italian company dedicated to the manufacture and personalization of handbags. For each bag, the customer must decide between several design options based on 3 different products. Once the model has been chosen, the design will be made in a CAD software. Originally, XYZBAG offers black as colour standard, although the customer can choose another special colour on request, which will be hand painted by professional craftsmen. For the creation of the accessories, the company uses SLS technology, although they recently integrated the one developed by HP, Multi Jet Fusion. For XYZBAG 3D printing provides great advantages to the fashion world, allowing greater freedom of design and customization.

Zap&Buj - The Spanish designers Elena Zapico and Raquel Buj, are the founders of the company Zap&Buj. The company was founded with the idea of combining architecture with fashion design, also integrating 3D printing technologies for the manufacture of clothing. To get an idea of the method they use, some of their designs were made with a flexible filament on an elastic tulle fabric. This allows the adaptation of the piece to the different body movements of the models. Both founders claim that additive manufacturing facilitated the materialization of organic geometries and the process of experimentation and prototyping prior to the creation of the sets. Zap&Buj already has several projects planned that, following the essence of the brand, will try to bring 3D printing, architecture and fashion closer together.

ZER Collection Design Studio - The ZER Collection design studio was created by Spanish artists Ane Castro and Núria Costa. Located in Barcelona, the company uses different digital manufacturing processes to develop its original garments. Their collection stands out for its use of environmentally sustainable materials. They usually use a 3D printer with flexible and biodegradable filaments to make the garments. They also have an industrial embroidery machine that allows them to create garments without seams or waste. Despite this, the studio is already thinking of integrating other technologies that will enable them to experiment with new designs, such as a laser cutting machine or a body scanner. ZER Collection brings added value to their garments, which is allowing them to reach a more varied public.

2.2 Advanced Materials for Creative Industries

Generally speaking, advanced materials refer to all those materials that represent advancements over 'traditional' materials. They may comprise new materials and modifications to existing materials that achieve superior performance in one or more characteristics critical for the application of interest in the creative industries. Superior performance does not only refer to technical or economic characteristics, but can capture

also other broader concepts such as sustainability. In fact, a deeper and more focused approach to advanced materials is to consider materials that are early in their product and / or technology life cycle. In other words, there is ample room for growth in terms of improving the performance characteristics (technology life cycle) and their sales volume (product life cycle). This selection is based on the latter definition and on the innovative capacity of the families of materials identified for the creative industries.

2.2.1 Advanced materials and its use in the creative industries

2.2.1.1 Active Materials

Active materials, also called 'smart', multi-functional or adaptive materials, are capable of modifying in a reversible and controllable manner any one of their particular properties whenever external physical or chemical stimuli operate on them. These materials have the capacity to change their colour, shape or viscosity, generate electricity, etc. in response to changes or alterations in the medium (light, sound, temperature, voltage). A first order classification of these types of materials include:

- Materials with shape memory;
- Electro active and magneto active materials;
- Phase-change materials;
- Photoactive materials;
- Chromo active materials.

Active materials have inherent or embedded "actuators" that respond to certain stimuli. They have response control and selection mechanisms, to control the response in a predetermined way. Their response time is short and the system returns to its original state as soon as the stimulus ceases. These materials can be used in the creative industries, with reference to the design of multifunctional products and can also configure intelligent structures and systems that, with a combination of materials, are able to self-diagnose and modify themselves to adapt to the conditions that have been set as optimal or correct for them.

CCI Focus – Active Materials are undoubtedly of great interest for the creative industries, just think of conductive fabrics for fashion, photo-luminescent or shape memory materials for furniture, sound-absorbing partitions. The design processes will be increasingly linked to how these materials can meet the possible needs of the population, the designer will therefore be committed to investigating the "training" process that will contribute to the exercise of the business.

2.2.1.2 Advanced composites

Composites are a broad family of materials. From a design point of view, it is worth regarding them by its structure, manufacture and behaviour, which render them different from monolithic or conventional materials. On the basis of these concepts, we can regard composites by one of the following features:

- It is manufactured artificially (thus excluding any natural materials such as wood), mixing the components in such a way that the dispersion of one material into another may be undertaken in a controlled manner to attain an optimal set of properties;
- It has two or more physically and/or chemically different phases or constituent parts, which are non-inter soluble and appropriately arranged and separated by a defined inter-phase.
- its properties are uniquely superior in one specific aspect and cannot be attained by its constituent components separately.

The development of composite materials is conditioned by several challenges such as:

- Reduction of manufacturing costs;
- Adaptation of automated manufacturing technologies to other industries;
- High standards of inspection and testing;
- The need for complete recyclability.

CCI Focus – Natural fibres composites (as lineo, jute, hemp) mixed with conventional material such as carbon or glass fibre improves significantly the damping properties of the material while ensuring good mechanical properties. Lots of seats and furniture projects are going to be realized through increasing knowledge of composites potential.

2.2.1.3 Advanced manufacturing;

Manufacturing processing of parts, although assisted by the most advanced controls, is still basically conventional: chipping, cold- or hot-forming, casting or injection. All of them face limitations such as the impossibility of curved drilling, collisions of tools with a part of complex geometry, restrictions in mould release angles to give just some examples. It is a barrier in the development of high value products with new functionalities.

The emerging of Additive Manufacturing with the advantage of all knowledge developed in the digital era is allowing to overcome such limitations. Is possible now manufacture through controlled deposition of material, layer by layer, putting down exclusively where it is needed to achieve the final sought-after geometry instead of stripping the material (mechanization, die-cutting, etc.); or shaping with the help of tools and moulds (casting, injection, folding, etc.).

The manufacturing processes of parts can be classified in the following way:

- Shaping technologies, which use pre-shapes to obtain the required geometry such as plastic and metal injection, PIM, sintering, vacuum casting, RIM, electroforming, etc.
- Subtractive technologies, which obtain the required geometry by subtracting material from a larger geometry such as mechanizing, electro-erosion, waterjet cutting, laser cutting, etc.
- Additive technologies (AddManu) which obtain the geometry by adding material through virtual geometry, without the use of pre-shapes and without subtracting material

A feature that distinguishes AddManu from other small-scale industrial manufacturing process is the easiness to produce complex geometric objects. This favours targeted product customization, as cost for the adapted process increases only slightly while results in unique

products that sell at higher prices. In addition to the cost advantages, the benefits lies in the design of new or differential products.

CCI Focus – Is possible to think advanced manufacturing in every field of creative industries as described in the previous paragraphs. Multi-material printing, printing on fabrics and reinforced/recycled/bio filaments are the most promising evolutions for home living and fashion.

2.2.1.4 Advanced textiles and fibres

In recent years, technical fabrics have undergone major development and offer many possibilities for innovation to create high-value products by offering new applications.

Until recently ‘sandwich’ structures were achieved by bonding polyurethane foam (PUR) to a variety of fabrics for the external layers. The development of 3D fabrics means that obtaining these sandwich fabrics is considerably simplified.

Currently, non-woven fabrics made by a mechanical punching technique or by means of needles are the most widely manufactured ones, as they have found many possibilities in the replacement of conventional fabrics. However, producing nano-fibres through electrospinning is gaining importance.

Electrospinning is a technique that allows to obtain fibres from molten or solution polymer with an average diameter in the range of 50 nm to 5 µm. There are studies in which electrospinning is used that have achieved composite nanoparticle materials in nanofibres or coaxial nanofibres. Recently a new method has been developed to facilitate the production of nanofibres called electrospinning based on polymer solutions or molten polymer.

CCI Focus – 3D fabrics have possible application in upholstered furniture, fashion and interiors because can be manufactured in a wide range of thickness, hardness, elasticity, with the advantage of being recyclable and can feature characteristics that are very similar to the polyurethane foam currently being used.

2.2.1.5 Coatings

Coatings play a prominent role for creative industries. They allow for transforming and/or modifying the functionality of a material through its surface, and in general with a metric economy that is worthy of note.

Nano-coatings are opening up new applications that are efficiently acting first-hand on the functionality of the material and the associated product. In this regard, nano-coatings are solid-liquid coatings comprised of extremely small particles that possess extraordinary characteristics such as: high flexibility, easy adherence, resistance to corrosion and microbial flora in addition to providing solutions for improving our environment:

- they save water by dispensing with excess cleaning;

- they provide protection and greater durability for materials, preventing their early breakdown, premature oxidation and molecular damage caused by harmful living microorganisms such as microbes, bacteria and viruses.

CCI Focus - Commercial examples of these nano-coatings for creative industries are for anti-graffiti, anti-corrosion, fire-resistant, anti-fungal, anti-friction, anti-grease and oils, anti-bacterial, self-cleaning, dry lubricants, self-releasing, polishing, photocatalytic application.

2.2.1.6 Gels and foams

Aerogels are the lightest solid materials known to man, as most of their structure is hollow. Their extraordinary porosity gives them a large surface area, which provides them with unique characteristics among solid materials. Their density ranges between 0.4 g/cm³ and 0.004 g/cm³ (only three times the density of air). This is because of their high porosity: more than 95% of their volume is occupied by air, giving rise to a high surface area. These characteristics give them unique properties in a solid material, like extremely low thermal conductivity and sound velocity and high optical transparency. They are considered to be the best thermal insulator, capable of withstanding temperatures of -50°C and melting at a temperature in excess of 1648°C. Recognized as the lightest solid in the world, it has a touch like foam.

CCI Focus - There are currently three crucial lines in the furniture sector and interior design solutions: the development of foams from recycled material and/or the recycling of polymer foams, such as polyurethane and polystyrene. The development of metallic foams made from aluminium, steel, lead and other metals with remarkable characteristics such as high stiffness, high resistance to compression and much lower density than non-foamed metal. The development of ceramic foams with density control. Some of the commonly seen applications are thermal and acoustic insulation, energy absorption systems, filling of metallic structures or sandwich panels in lightweight structures and development of ceramic foams with density control.

2.2.1.7 High-performance polymers

Modification and reinforcement of compostable, degradable and/or conventional polymers (or a mixture of them) with bio-fibres and/or nano-charges can result in materials with very advanced properties for innovative applications.

Traditionally, the use of charges with polymers has had the purpose of reducing the product's cost and of improving its physical-chemical properties. Charges are normally small particles, short fibres, organic or inorganic materials. The main advantages of using organic fibres to reinforce plastics are: they have low densities, they are non-abrasive, easily recyclable, biodegradable with a low energy consumption and low cost.

CCI Focus – High performance polymers are used to produce ergonomics and flexible chairs, advanced shelf solutions, technical parts of sports shoes.

2.2.1.8 Light alloys

The creative industry is constantly searching for light, strong materials that can be used in a large number of goods. Its objective is to develop structural materials with high specific strength, which is the resistance or strength of a material divided by its density. The following

four non-ferrous metals are the most used lightweight metals in CCI: aluminium, titanium, magnesium and beryllium.

CCI Focus – Light alloys are used to produce sitting frames, tables, profiles for the kitchen industry, heels for high-end shoes, or eyewear sector.

2.2.1.9 Nanotechnology

Nanotechnology refers to a comprehensive field of applied science and technology whose unifying theme is the control of matter at molecular level, which is smaller than a micrometer, normally on scales of 1 to 100 nanometers. It includes the manufacture of devices on a nano-scale. It is a highly multidisciplinary field seen across applied physics, material science, colloidal science, supra-molecular chemistry and electromechanical engineering. Nanotechnology can be considered as an extension of sciences on a nano-scale. Two main approaches are used in nanotechnology: top-down and bottom-up. One of them is bottom-up, where materials and devices are built from molecular components, mostly in a self-constructing matter. In the top-down approach nano-structures are built by using techniques like photolithography or nano-imprinting.

Carbon nanotube was the pioneering material in this technology and today graphene is the most researched for applications. Carbon nanotubes are based on cylindrical nanostructures made from carbon atoms. It is renowned for its unusual resistance and capacity to conduct heat and electricity. Graphene, in turn, is transparent, flexible, extraordinarily resistant, impermeable, abundant and conducts electricity better than any other known metal. This material permits manufacturing of electronic devices with flexible and transparent screens and ultra-rapid batteries, powerful solar panels, applications in aeronautics, medicine etc. Nanotechnology is an excellent base for creating new materials, according to specific needs. It is also a source of inspiration for other two-dimensional materials such as fluorographene, a two-dimensional analogue of TeflonTM with extraordinary lubricating and insulating properties, hexagonal boron nitride, a very hard crystalline and transparent insulator that combined with graphene improves its electromechanical properties, molybdenum disulfide, another two-dimensional crystal with promising properties for the construction of a new class of transistors or silicene, a version of graphene made from silicon that can be easily integrated with current silicon-based electronics.

CCI Focus – Nanotechnologies are going to be used more and more in textile industry, fashion products and advanced surfaces or covering solutions for home living.

3 Emerging Trends affecting CCI

3.1 Sustainability

Consumption patterns have evolved over the past decade in Europe and so has also the European creative industries sector. In particular fashion is experiencing a period of changes, one of the trends being a demand by clients for more sustainable production and consumption models. At the same time, the spread of technological startups strongly focused on the radical transformation of the supply chain and international logistics, as well as on the use of materials with low environmental impact, is accelerating the introduction and sale on the market of compatible solutions from the environmental point of view.

In the fashion industry, the biggest problem lies in the volume of unwanted clothing that ends up in landfills. According to the Ellen MacArthur Foundation [AA.VV, 2017]⁹, which works to promote sustainability, global apparel production nearly doubled from 2000 to 2015. Over the same period, the number of times a garment was worn fell by 36%. All in all, "the equivalent of a garbage truck full of clothes is burned or dumped in a landfill every second," their report noted. Around the same time, according to the World Economic Forum¹⁰, 60 % more garments were bought, but consumers only kept them half the time.

Sustainability can be analysed within the five broad phases of the life cycle of a product (pre-production, production, distribution, use and disposal), leaving individual responsibilities to producers or consumers, or evaluating, as always more often it happens, collaborative approaches in the management of the end of life.

End of Life

Some companies are trying to increase their sustainability by encouraging consumers to activate virtuous recovery circuits. There are various strategies to raise awareness on the value of the garment or furniture. In some stores, for example, customers may pay a nominal amount to transform unwanted clothing into new ones through a process that breaks down old fibres and combines them with new ones. Other companies seeking to provide "cradle to grave" solutions, an expression used for processes that intend to keep materials within a circular economy by having set the recovery and recycling dynamics right from the start of the design process. Therefore, digital and non-digitized services are being tested to actively collect products, also collaborating with important players in the shipping sector. The main objective is to structure sustainable recovery and recycling models, often 'bypassing' providers of environmental services or recyclers of urban and industrial waste.

⁹ AA.VV., (2017). "A new textiles economy: redesigning fashion's future" - Charity Registration No.: 1130306 OSCR Registration No.: SC043120 Company No.: 6897785".

¹⁰ <https://www.weforum.org/agenda/2020/01/fashion-industry-carbon-unsustainable-environment-pollution/>

Box 4 - Empty your Wardrobe

The vision of Empty Your Wardrobe (EYW) is to create a platform for sustainable innovation, connecting people from different backgrounds all focusing on both minimizing textile waste and creating a sustainable creative hub for opportunities. EYW reduces textile waste by extending the life of used clothing and accessories following the circular economy doctrine; reduce, reuse, recycle. The project also aims to empower communities, corporations, initiatives and individuals to create, establish or launch their own creative and sustainable ideas, products, initiatives or services

Recycling

The ongoing challenges essentially concern mechanical and chemical recycling approaches. If the first is the most widespread, it is quite evident that it continues to have many limitations, according to industry experts deriving from the use of less advanced equipment compared to the complexity of the processes and the quality required in the selection processes. Chemical recycling, on the other hand, seeks to reduce fibres to their basic components and rebuild them with less impact. Some cottons, for example, can be converted into cellulose fibre, currently produced only from wood. Recycling processes require advanced sorting, classification and shredding systems, as well as identification tools (for example cameras) to intercept the composition of a fabric with greater precision. Once selected and chopped, the fabrics can be dissolved and transformed into pulp, re-polymerised and transformed into new semi-finished products.

Box 5 - Castor Sunglasses

Castor Sunglasses is a brand that offer glasses with a radical and quality design made only from recycled and ecological materials at a revolutionary price. The business is socially conscious and they intend to reinvent the eyewear business by offering a quality product made only with recycled materials. Castor, for example, works with wood products that are always linked to sustainability, but working with any type of wood and the means with which they are worked do not always guarantee a sustainable result. They strive to make the whole process as ecological as possible, making recycling a small obsession and taking care of important details such as the sustainability of wood and the non-use during the manufacturing process of toxic and / or aggressive products with environment. Castor have further several collection based on recycled materials, as recycled tyres or bio acetate.

Veganism

Another area of interest concerns new fibres and materials that are based on products found in nature but not derived from animals. Several companies, for example, are developing alternatives to leather, as hides are particularly problematic, from the methane-producing cows that produce them to tanning methods that often involve toxic chemicals like chromium. Many eco-leathers, despite the prefix, derive from plastics and for this reason they do not solve the problem, but simply move them. Some alternatives involve mushroom skin, which relies on mycelium, or mushroom roots, to produce an animal-free alternative. Today there are prototypes of naturally dyed sports shoes grown with microbes (after a week of composting they reduce their size and lose more than 70% of their mass), as well as alternatives to wool fibre created through biodegradable fibres based on the DNA of already existing proteins in nature.

Box 6 – MycoTEX

MycoTEX is a ground breaking seamless manufacturing method allowing for custom-made clothes made from compostable mushroom roots. They use a seamless production technology to create custom-fit products out of sustainable, vegan textiles made from mycelium (mushroom roots). This ground breaking manufacturing method solves several major issues in the fashion and interior industry: reduces the cost, waste, and labour-intensity of cut & sew operations; replaces plastics and leathers with compostable materials; improves the comfort and fit of fashion products.

In the furniture sector, sustainability seems to be slightly behind fashion (driven by the increasing sensitivity in the purchasing processes of the new generations). The main environmental performances are based on energy consumption, climate-altering emissions and waste reduction. In terms of investments in Research & Development in this sector, Italy leads the ranking with 56.4 million euros, ahead of British (44.6), German (39.9) and French (17.5) companies. The main investments concern the reduction of energy and water requirements and the elimination of toxic substances, also according to the parameters required by large companies and the public administration. But one of the most important trends concerns the change of strategy on logistics and reverse logistics (i.e. the recovery and collection of returns). The sectors most affected are those of seating, kitchens and office furniture in general. Among the semi-finished products at the centre of the transformations of the green economy, the percentage of recycled wood in the production of particleboard (which still forms the basis for furniture products) is growing. Another trend concerns the study of truly sustainable paints, considering that in many cases paints with water they do not have the same effects as traditional ones and in any case most of them are not biodegradable.

The concept of sustainability, which is purely productive, is accompanied by more and more business models that replace the concept of "end of life" through the reduction, regeneration and recovery of materials in production. This path is aimed at saving resources, but gradually translates into creative recovery approaches by providing new ways of imagining objects both in the short and long term, creating micro systems of industrial symbiosis. Thus, the growing activation of networks that help SMEs to decline or support these models can be seen.

3.2 Product and Service Differentiation

The CCI, considering their nature and their main vocation, that is to elaborate and direct creativity towards recognizable and attractive products and services, are the ones most involved in differentiation processes, a fairly extensive concept that can be analysed from different points of view. Product differentiation, at least in general terms, is what distinguishes the product or service from a market characterized by a comparable offer. For this reason it can also be understood as the perception that a potential recipient has of the product, aimed at undertaking cognitive, exploratory, purchase and loyalty actions.

Digitalization and new technologies have allowed creative industries to explode some possibilities and to inaugurate differentiation strategies based on specific factors, such as the level of personalization and the interaction experience for the achievement of goods and

services with unique characteristics or performances or compatible with purely subjective qualitative elements.

We therefore moved from a focus on the aspects of style, production, quality and sales, to a focus based on how the potential customer would imagine a good with respect to specific needs of natural anthropometric, aesthetic and functional, therefore, with respect to needs related to the target person or environment. The simplification and optimization of the personalization experience are now considered the key to understanding and the main objective to ensure that the technical potential is translated into an object with the desired characteristics. And if by simplification we mean the speed with which a customization process can be concluded with satisfaction, by optimization we can multiply the field by including tools that allow an evaluation of the asset beyond its intrinsic characteristics. In the fashion sector, for example, I can customize a colour based on personal taste. But if technology allows me to simulate the colour of my skin and evaluate a garment worn also in relation to the performance on the skin, the experience can be considered extended or increased. The same goes for a table that can be customized in terms of wood surfaces and grain, to which I associate a possibility of contextualization in a real, virtual or mixed environment (real + virtual). This type of approach tends to lead the customer to recognize a real approach of the CCI towards his needs, with a consequent greater willingness to purchase products that offer a unique and useful experience.

It is important to differentiate products in any industry, but especially if you are in a crowded market with many competitors as creative sector. The goal is to show potential customers what company can offer that other businesses can't and why it's valuable to them.

3.2.1 Types of Classification and their Competitive Advantage

Product or service differentiation is an asset as a competitive advantage. The creative industries, today, are measured mainly with three types of competitive advantage: benefits, approach to the recipient and competition.

Benefits are the values a customer receives when purchasing a product or service. They are different from the aesthetic characteristics or functionality, which are the characteristics attributable to the primary function of the asset. Creative companies are focusing more and more on the benefits related to the purchase, expanding the concept of value from the quality / price ratio to the ability to establish benefits extended beyond the purchase or in some cases in the medium and long term.

The recipients of the product, once classified almost exclusively as customers, today represent a pool of subjects through which to sell, but also to develop or improve the product itself. The creative industries therefore fuel a gradual transformation of the passive to active approach, where the differentiated advantages of the product are aligned with the interests, needs and values of a defined target audience. Differentiation has widened the field of desire from "why he wants it and how much he will pay" to "how he wants it to appear and where he wants to buy it".

Finally, competition helps to understand what the margins of differentiation are and which strategies have already been probed and applied. The value of the creative product lies more

and more in the perception of how the advantages and specific needs of the potential buyer are distinguishable or alternative to the solutions populated by the competition.

In this study we distinguish five types of differentiation, three of which are traditional and two that we can call “advanced”.

- **Vertical differentiation:** occurs when customers choose a product by ranking their options from best to worst using an objective measurement, such as price or quality.
- **Horizontal differentiation:** occurs when customers choose a product based on subjective preferences, because they do not have an objective measurement to distinguish between the best or the worst;
- **Mixed Differentiation:** Customers who make more complex purchases tend to use a mix of vertical and horizontal differentiation when making purchasing decisions. Many creative products, especially with the advent of the IoT, select vertically for functionality and horizontally for aesthetics;
- **Extra-combined differentiation:** it is a type of differentiation of a mixed type, but guided by extra criteria concerning the customer's ability to marry the specific purposes of the producer / service manager, such as sustainability, ethical production, support of disadvantaged categories, solutions to problems of a global and non-particular nature;
- **Selective differentiation:** customers are grouped numerically, the differentiation field is strongly pushed and therefore focused on lead users or experts. It is based on how the product is sold and managed over time, on the quality of the service, on the duration, on the adaptation or upgrade of functionality.

3.2.2 Custom versus Bespoke Design

European Commission¹¹ has recognised the degree of digitalization of companies as one of the key enablers of innovation. In the design sector digitalization has greatly contributed to increase the involvement of customers.¹² There are, however, different levels of involvements and these have been subject to investigation. In particular, the census of the CCIs revealed various customer involvement strategies identified of which it is worth introducing the concepts of ‘custom’ and ‘bespoke’ design and highlight the differences.

- **Custom design** starts with a pre-existing template or model that can be adapted to the needs of an individual only within pre-defined limits. For fashion and furniture, for example, customers can select from pre-existing options, such as materials, colours, the positioning of accessory elements, or other pre-defined choices. In product design, companies provide configurators within which customers can select a range of options, while in the service design sector generally templates be populated with content. In all cases, however, it is always the company that delimits the fields and choice of

¹¹ European Commission - Monitoring the Digital Economy & Society 2016 – 2021, prepared by the European Commission; DG Communications Networks, Content & Technology, Bruxelles (Belgium), 2015

¹² While technologies like 3D printing clearly rely on digital, others such as new materials may less (or not) be related to digital at all. To which extent digital technology plays a role in various examples of AddManu presented above, is outside the scope of this report.

In this paragraph you refer to the role of digital in the link between producers and consumers. That can often be separated from the role of digital in production.

customization. Once the choice has been made and has been purchased, it is no longer possible for customers to intervene.

Box 7 – Mobibam

In 2016 Mobibam launched a concept to sell personalized furniture online. Their web-based configurator allows clients to design and purchase a piece of furniture fitting exactly their room size. The 'sizing' starts by applying library of solutions to optimize the space available in the apartments and to adapt to the subjective wishes of the customer with respect to size, aesthetics and functionality. The latter no longer passes by a salesman, he no longer passes by a designer to whom he would delegate his project, it is he who commands, it is he who decides everything, with his time and from his home. By eliminating the intermediation of a professional in charge of surveying and designing the layout and delivering a piece of furniture to be assembled at the home with relative instructions, the company realizes the custom-made furniture with competitive methods compared to traditional artisan productions.

- Contrary to custom design, in **bespoke design** products are "made-to-measure" or "made-to-order". This flexibility leads to a completely original product or service that responds to specific non-standard needs of the consumer. In fashion, for example, anthropometric scanning systems are employed to capture the client's measures without having to introduce data in the product's configurators. Moreover, it is always the consumer who directs the personalization field and the personalization process often continues *in itinere* until delivery.

Box 8 – Iris van Herpen / Ganit Goldstein

Iris van Herpen Haute Couture atelier is located in Amsterdam and is a laboratory that blends new ideas with new techniques, using a team of graphics, tailoring, embroidery and material development. The atelier specializes in laser cutting and 3D printing to develop new geometries or functional drapes for the creation of customized dresses with sophisticated silhouettes, using meticulously selected fabrics and refined technological aspects that seek to set new standards. The atelier is engaged in research and development activities, such as in the application of ultrasonic welding to create seamless seams, using acoustic vibrations and high tones of sound to connect the molecules of the material; or in the development of innovative mixtures of magnetic powder and liquid silicone used to model 3D textures. She has designed a nature-focused collection that included the Foliage dress with 3D printed sheet-like patterns. Using a PolyJet 3D printer, Iris van Herpen was able to create 0.8 mm thick sheets based on tulle. The designer has also created a number of 3D printed shoes in collaboration with the United Nude brand. The Haute Couture process is based on a creative consultation to understand needs, preferences and desires, after this they work on a selection with options that vary in color, form, details and materials.

The stylist Ganit Goldstein has been using 3D printing for several years now to create custom-made clothes that are perfectly adapted to each individual. Her first collection, called “Between the Layers“, includes 7 garments and 6 pairs of shoes printed in 3D on Stratasys machines, including a PolyJet that allows to combine multiple colours. Ganit Goldstein explains that she always starts her work with a 3D body scan to adjust her designs to a specific silhouette. In this way, she hopes to offer clothing that is personalized, comfortable and adapted to each individual.

3.3 CCIs as a revived opportunity for the development of rural areas

The emerging trends analysed in the previous chapters, especially as regards additive manufacturing and the production of advanced materials in geographic areas characterized by the high presence of secondary raw materials and waste from the agro-industry or the extractive industry, open up scenarios of great interest in the creation and strengthening of creative industries in the peripheries of Europe, or in contexts circumscribed at regional level characterized by a low population, few companies and large distance to large urban centres or industrial areas.

A necessary premise concerns the direct and indirect needs or benefits that could arise from this scenario. Creative workers have the flexibility to work from anywhere as long as facilities are available, especially a quality broadband connection. Preserving the quality of life, landscape, built environment and natural heritage of a rural region appears to be key to maintaining its attractiveness to people and creative businesses. Opportunities for social, cultural and commercial interaction between creative professionals can also play an important role in rural regions where isolation can be a problem and where there is a risk that creative talent may be lost in favour of larger centres. Another aspect concerns connectivity, both physical and virtual, which is fundamental for the functioning of creative businesses in rural areas. Currently, many of the creative sectors at the regional level mainly serve local or national markets; it is likely that the future growth of the sector will require an expansion of the export business, with greater connectivity key for this. There is a possibility, however, for tensions to arise if greater connectivity in rural regions impacts their 'creative place' attributes. Finally, there is an ever greater reduction in public resources to support the sector, with particular reference to the artistic departments of local authorities, investment funds for cultural infrastructures and other sources of financing for the creative sector.

The importance of environmental factors and quality of life is therefore fundamental to retain and attract creative talent in a rural region, as well as overcoming some constraints of a rural location to the functioning of creative enterprises, particularly in the infrastructure sectors, accessibility and networking. Digital creative factories could be the appropriate and evolved answer, both in the form of small digital workshops and hubs of experts in the use of ICT technologies for the creative industries. Digital creative factories are, to all intents and purposes, small-scale factories equipped with laboratory or industrial-grade equipment, electronic tools and software used for the production of prototypes or small series. The digital creative factories arouse great interest at the local level, just imagine for example the islands or the extreme provinces of the continent, since they can concentrate resources and experts towards creative purposes, or towards the development of new manufacturing and tourism products, towards repair and maintenance of goods produced with industrial production

technologies, towards solutions that in some cases help small towns to express their own production culture that is increasingly distinguishable.

Despite technological and infrastructural advances, several regions of Europe suffer from the low concentration of companies, the distance from supply chains, the increasing comparison with labor costs in non-EU countries. Creative businesses operating locally, therefore, express strong values in terms of craftsmanship, but do not have tools to intercept interests from richer areas or to aggregate skills in order to integrate specific know-how with the use of local materials in a digital key, or, above all, on increasingly pushed and customized design elaborations.

In the business models that we will analyse later, two main profiles emerge that could contribute in an important way to the development of rural areas:

- **Providers of novel materials for CCIs:** manufacturers of filaments, spools, profiles, sheets and semi-finished products in general, which can be processed through 3D printers and CNC machines. These micro-productions would be of interest if associated with the ability to collect, select, shred and mix interesting quantities of post-consumer materials or agro-industry waste, developing compounds suitable for specific applications;
- **Digital Artisans** able to manage the whole design service based on analyse, process and transform product files into fashion accessories or goods for home living.

These profiles would contribute to more ample societal challenges, such as:

- Fighting against the depopulation of rural areas, avoiding migration to urban productive areas;
- Integrating specific local knowledge with the opportunities deriving from new technologies;
- Recovering of waste and scraps as new resources, activation of circular economy practices;
- Creating of training centres (design, modelling, manufacturing) and assistance for on-demand production or for self-productions;
- Increasing employment and economic growth of the contexts in which digital creative factories operate.

Box 9 - MareTech

The MARETECH project re-evaluates the sedimentary materials discarded from the French Atlantic coast in the field of design using 3D printing technology. Thus, today, on the French Atlantic coast, natural materials are created through sedimentary processes in the water and reach the seashore through the tides. These waste materials are usually collected and thrown offshore. Designer Aurore Piette has devised new reuses and manufacturing methods to provide design applications to these raw materials, inspired by craftsmanship and nature. To date, prototyping activities are underway for modular architectural elements to configure partition walls for interiors. Aurore Piette's goal is to develop a network of collection and processing points, training new digital artisans of the sea and contributing to the creation of micro-economies based on the enhancement of local natural resources and the use of emerging technologies. MARETECH could therefore become a strategic training and design hub capable of identifying rural and peripheral areas where to set up mini additive laboratories, zero waste processes, and transformation of sedimentary materials into design projects that embody the cultural, environmental and geographical values of areas in which they are made.

4 Analysing the Industry Landscape

4.1 Mapping Innovative CCI Companies

The authors carried out a desk research to retrieve innovative companies in the CCI sector located in the EU. It is worth mentioning that practically the totality of these companies are small companies and some of them mid-sized. Therefore, these companies cannot be retrieved by just querying standard enterprise such as the Moody's BvD Orbis database, nor there is a suitable list, to our knowledge.

The first step the authors was to establish a list based on different sources. The first source, were CCI startups; for which the authors accessed the commercial databases Dealroom.co and CBinsights. The second source, were CCI companies that had participated in H2020 programmes, such as the WORTH programme. The third source were company's participation in design congresses, workshops or other events. Finally, the list was completed by web search.

The companies retrieved, were then scrutinized one by one to check whether and to which extent they complied with the innovation criterion, i.e. using additive manufacturing tools or employing advanced materials. This way we created a database with 430 companies.

The database contains basic information on the firm, such as the business name; location / country, website; and contact email. Specifically for purposes of the research, we populated as follows:

- Type of Business: [informal team, company, design studio, fablab, association]
- Typo of Company [SME, Large Enterprise, Startup]
- Main Industry and sub-industry [sector in which they generate the largest (2nd largest) share of turnover or primary (secondary) interest];
- Client Focus: [B2B, B2C, B2B / B2C Hybrid Model]
- Custom Services (bespoke design, custom design/online customization services);
- The main Focus on production / specialization (Material Innovation / Sustainable Materials, Smart Textile / e-Textile / Wearable Technologies, Digital Services - AR/VR/AI/IoT/Multimedia, Additive Manufacturing Services / Additive Manufacturing Production, Generative Design / Parametric Design);
- Advanced materials used in design, prototyping and production activities;
- Degree of digitalization;
- Innovation [use of Additive Manufacturing as process or for final products; use of advanced or sustainable materials for final products; etc.]
- Private Label: [whether they market and distribute under their own brand];
- In addition for Startups
 - Launch Year
 - Development Stage: [Seed, Early Stage, Growth (Later stage)]
 - Funding received by venture capital, business angels, government

In addition we collected any piece of information that may indicate novel application routes, ingenious approaches, inspiring outlooks, forward looking insights, etc.

4.2 Mapping by CCI Sectors

The database assigns the companies to a CCI sector. For the purpose of this study we have found that the following ten sectors properly describe the realm of the business models of the companies to be analysed.

1. Fashion
including Textile, Clothing, Footwear and Jewellery/Accessories
2. Fashion Design
Includes design services for TCF and Jewellery/Accessories. Note that the producing products (fashion) and selling fashion design are distinct business models. Companies may serve one or both of them
3. Furniture
4. Furniture Design
Relates exclusively to the design services for furniture. This is similar to point 2.
5. Home Living
Accessories and furniture items which are easy to replace and easy to move
6. Interior Design
7. Material Provider
Laboratory or SMEs engaged in production of semi-finished for creative industries.
8. Service Design
Configuration of projects and programmes to empower creative industries results and create new customers and users experience
9. Technology Provider
Professional services designed to facilitate the use of technology by enterprises and end users, creating smart and interconnected objects, digital interaction, virtual and augmented reality.
10. Marketplace
Online platform specialized in selling/delivery or customization services.

4.3 Digital Creative Industries

Information and Communications Technology (ICT) is a general purpose technology that brings additional innovation to the organization, businesses and industrial sectors that adopt it. Therefore, the digitalization of the creative industry sector is closely connected to degree of adoption of ICT by the individual companies. In particular, the degree of integration of ICT

within the company organization and internal processes and can be characterized along the following three dimensions (European Commission, 2016)¹³:

- **Connectivity of the CCIs:** connectivity, inside and outside the proprietary site, is an important precondition for participating in the Digital Economy and access to the Internet is increasingly essential. At the same time, Digitalization requires the recruitment of individuals who possess the necessary skills to use the respective technologies. In view of the marginal relevance of connectivity for the purposes of this study, it will no longer be taken into consideration;
- **Internal processes** of the CCIs: In creative companies, the integration of Digital Technologies to manage internal processes or the offer of innovative products, which represent the digital technologies themselves, are fundamental conditions for the growth and differentiation of the offer. According to the monitoring work carried out by the European Commission, we can extrapolate three distinct levels:
 - Digitalization of production processes (Additive Manufacturing);
 - Electronics or sensors incorporated in the products (IoT);
 - Change of business models.
- **Customer involvement:** The Digitalization of the economy also concerns the change in the way companies involve their customers in business activities. More and more companies use websites with e-Commerce functions or social media to carry out online Marketing and Advertise actions.

To observe how the digitalization of businesses is influencing the world of European CCIs, it is possible to extrapolate from the database digital companies, distinguished from non-digital ones through the previous degree of Digitalization. The selection of digital start-ups involves all those companies that have highlighted the existence of at least one degree of Digitalization between "Internal Processes" and "Customer Engagement".

a) Digital Creative Industries as Digital Technology Agents.

The company offers on the market products or services that enable Digitalization, as they represent the enabling Digital Technologies themselves. Proposing products or services enabling Digitalization also implies the consequent integration of Digital Technologies in the same internal company processes. The Digital Technologies offered and considered enabling are the following:

- Platform: offer of a Service, intended as the development and maintenance of a proprietary multiside platform;
- Proprietary or on-order software: offer of an Intangible Product, intended as the development of proprietary software, such as APP or management applications, or for third parties on order of any kind.
- Virtual Reality (VR): offer of an Intangible Product, intended as the development of 3D graphics, application for Augmented Reality and Virtual Reality solutions.

¹³ European Commission "Monitoring the Digital Economy & Society 2016 - 2021" (2016)

- Bionics: offer of a Material Product, intended as the development of bionic devices for the exoskeleton.
- Multimedia Contents: offer of an Intangible Product, intended as video production and / or post-production and digital multimedia content of all kinds.
- Artificial Intelligence (AI): offer of an Intangible Product, understood as the development of solutions that enable AI.
- Internet of Things (IoT): offer of Material Product, understood as the development of systems that exploit IoT technology.

b) Digital Technologies in internal processes.

Digital Creative Industries because they are the User of Digital Technologies in internal processes. The creative company offers products or services on the market that are not considered enabling Digitalization, but the integration of Digital Technologies into internal production processes is evident. Enabled Digitalization can be observed in the following internal business processes:

- Manufacturing: design and production of Material Products;
- Design: offer of consulting services for virtual design (3D, CAD, ...) and possible 3D printing of physical prototypes;
- Web Marketing: offer of a consulting service for corporate marketing on the Web and through the use of social networks.

c) Digital Creative Industries for Customer Engagement.

The creative, digital enterprise is limited only to the active involvement of the customer within the production process, with the possibility of customization, choice or booking. It can be observed that a start-up is subject to this Degree of Digitalization if it meets only the following requirements:

- Presence of a web-page with an e-Commerce function, proprietary or supported by an intermediary platform;
- Advertise online and use of social media for Marketing strategies.

4.3.1 New business models

From the analysis of the CCIs surveyed and their degree of digitalization it can be seen that traditional business models and advanced models still exist, the latter strongly influenced by the Digital Economy and the centrality of the customer in the products and services offered. The development of new business ideas that use digital technologies are mainly customer-oriented and the proposal of new and disruptive business models tends to favour the User-Intensive development approach, which orients the functionalities of the proposed product or service based on the analysis of needs and interaction with users, rather than an exclusively Technology-Push approach, which instead puts the proposed technology at the centre of development. It should be emphasized that, as a consequence of the Digital Transformation of society, it becomes increasingly important for creative companies to interact with their

reference Community, or in general with the target market, even during the product development phase and before the marketing phase, anticipating the phase of communication (marketing) with the market.

The spread of digital infrastructures and the emergence of increasingly complex but simple to use digital technologies have together determined the evolution of the business models proposed by creative companies. To rank some of the most interesting models, we considered the elementary framework of Baden-Fuller (2016) as the preferred approach for this study. The four models can be used to identify, classify and analyse business models in this research. The four elementary models are:

- Product model;
- Solutions model;
- Matchmaking model;
- Multi-sided model.

The product model involves the sale or licensing of goods and services; as described before in this paper, it is the most common model found in the CCI: “The company develops a product or standardised service and sells it to customers. The value proposition is transactional: to provide a product or standardised service that customers will buy;

Solutions models are more tailored, and are driven by the firm’s response to consumer needs, such as a consulting firm: “The company engages with a customer about a problem the customer faces, and provides an integrated solution. The value proposition is relational: to tailor solutions to each customer;

Matchmaking involves the firm matching buyers and sellers within its marketplace; the creative company joins customers and sellers in its online or physical marketplace. The value proposition is transactional: to facilitate exchange.

Multi-sided involves the creative offering products or services to different consumers, and these consumers derive additional value from each other. The company provides different products or services to different customer, designer or manufacturer groups. The value proposition is multi-sided: one customer group gets additional benefits from the other group’s transactions.”

We can classify product and solutions models as dyadic, indicating that they are dominated by relationships between two parties (company-customer.) However, the matchmaking and multi-sided model are instead triadic or more, incorporating both the firm’s relationship with consumers, and the relationship between those customers. In triadic models we can observe competition and cooperation phenomena that generate further external relationship among the actors.

4.3.1.1 Matchmaking

At least two creative subjects are present in Matchmaking models: the one who generates the contents and the other who hosts them. By connecting to the previous definitions, we can talk about a Digital Creative Agent (who develops and publishes the platform) and a Digital Creative Designer (who publishes the contents). Customer can introduce creative components providing ideas and models as an integral part of larger projects (a practice in use especially in the world of architectural design). Designers can, for example:

- sell 3D Models for Computer Graphics (CG);
- sell 3D Models for Additive Manufacturing (AddManu);
- sell Concept or engineered projects;
- sell physical products;
- sell DIY projects and instructions.

The opportunities come from:

- Mega Traffic;
- High Visibility;
- Promotional banner and publication on newsletter;
- international customers.

Matchmaking models are usually royalty based:

- Up to 90% of selling on some platforms;
- Higher royalties if the collaboration is exclusive;
- Designer can set their markup;
- Fixed entry commission.

The main reflections and research ideas deriving from these models relate to the fact that they are often quantitative approaches, based more on the number of published ideas and transactions than on the quality of the projects. There are several possibilities of parallel services deriving from the exchange if, in the first instance, suitable or adaptable models are not available on the platform. It should also be noted that many projects are often shared for free (in this business model the goal is to increase the number of users and the collection of files in order to increase traffic) with revenues deriving from direct-marketing or advertising strategies.

4.3.1.2 Multi-sided

At least three creative subjects are present in the Multi-Sided models: who generates the contents, who hosts them and who produces them. By connecting to the previous definitions, therefore, we will have a Digital Creative Agent (who develops and publishes the platform), a Digital Creative Designer (who develops and publishes the contents) and a Digital Creative Manufacturer (who has compatible technologies for processing and production the contents in a physical way), in turn linked to a Material Provider. Designers and manufacturers can, for example:

- sell 3D Models for AddManu or CNC production and relative instructable (packing, assembly, painting, recycling);
- AM or CNC companies can produce and delivery at local level;

The opportunities come from:

- Custom products based on dimensional control and several material, texture, color;
- On-demand production;
- Collaboration between Manufacturers, Material Providers and Designers.

Multi-sided models are usually based on revenue that come from:

- Designer set their markup on the download of digital model;
- Platform set entry commission for each designer and AddManu/CNC fabricator;
- AddManu/CNC fabricator calculate in terms of mass, volume, material and manufacturing time the final price.

The reflections and the main research ideas deriving from these models lead to imagine the development of Creative Digital Factories (an aspect that reconnects us to the development of rural areas). These models based on pushed customization are driven by a qualitative approach for high-spending markets. The supplying of 3D printing materials or semi-finished products is almost always functional to small productions, waste reduction and local procurement. Customization on several levels leads to altering and certainly making the aspects of IP considerations / threats more complex regarding digital distribution of Design. An aspect of great interest concerns the possibility for manufacturers to make optional the use of their tangible and perhaps unused assets (equipment with low productivity in certain seasons, non-working hours, semi-finished warehouse already subjected to cutting processes, etc.) The revenues would derive from the payment of the sharing services that can be activated from the platform

4.3.2 Consumer as producers, as integrator of resources and as remixer

Consumers are increasingly theorised as producers on their own or through participation with firms and/or other consumers especially in online communities. From a theoretical perspective, consumer productivity is evident in many strategies and visionary trajectories of creative industries. On this basis is important to develop a better understanding of 'prosumption and the emerging role of the prosumer.

Consumers actively involved in the practice of consumption become prosumers for a variety of reasons, not least the creative attitude or the need to modify / improve products that are part of their daily life. This means that, by becoming a prosumer, one is asked to be more involved in consumer experiences rather than relying on commercial suppliers or becoming a producer who establishes co-creation dynamics. New technologies, therefore, in addition to bringing machines into homes, are exercising practices that concern, more than mere collaboration, consumer empowerment as prosumers acquire more self-dependence (or self-sufficiency) for products or services that otherwise they would acquire as such on the market. A not secondary aspect is also the reduction of susceptibility with respect to the exploitation imposed by the supply chain, since the prosumers relate to the dependence on secondary intermediation (for example agents or retailers) and commercial offers. What certainly

appears interesting in the survey is that Prosumers intervene in the strategies of the creative industries at different levels and which, gradually, as designers and producers, are increasingly establishing themselves as integrators of resources. This is with the vision of manufacturing products as the results of a broad integration of resources such as people, technologies, raw materials, knowledge and information, which a company bears entirely and for which often excessive, superfluous and recurring costs are incurred.

If on the one hand, therefore, we imagine the prosumer as engaged in tangible aspects, from co-design to the self-production of goods and components, it is interesting to imagine the consumer as a key player in a collaborative path based on the following categories:

- human resources, supplier of knowledge to the creative enterprise;
- organizational resources, provider of data on use, wear, management and reliability;
- information resources, provider of knowledge in terms of market segments, competitors and technologies;
- relational resources, supplier of relationships with local consumer groups, small distribution.

Finally, it must be considered that new technologies and their use within increasingly advanced platforms offer unprecedented flexibility in adapting existing product designs to changing needs. Prosumers, to benefit from this potential, are increasingly equipping themselves with structured design procedures and tools, while online platforms for creating collaborative projects are implementing more and more features for remixing, a concept that describes the creation of new models based on existing design elements. The value of remixing lies in improving product design processes, whether they are oriented towards creativity (inspiration, learning) or productivity (speed, improvement). The remix helps to increase the quantity and diversity of projects, it is a fact that projects created by remixers are printed and produced much more often by community members, suggesting that the remix helps ensure compatibility of production similar to improvement continuous process.

4.4 Emerging Trends

In addition to the desk research on 430 European companies, which fulfilled the original database, the authors complemented its research by sending a questionnaire to these companies. 83 questionnaires were received, permitting to investigate emerging trends detailed in paragraph 4.4.1, 4.4.2 and 4.4.3.

4.4.1 Survey results

Out of 83 companies surveyed, 52 declare that they intervene in a secondary sector. This means that many creative companies develop parallel ideas, services and collaborations, with a high degree of diversification. In the fashion sector, for example, manufacturing companies are engaged as a secondary activity in stylistic design or in the supply of materials.

The distinction between material provider and technology provider is very clear, only one company has declared to provide advanced materials and technologies for its implementation.

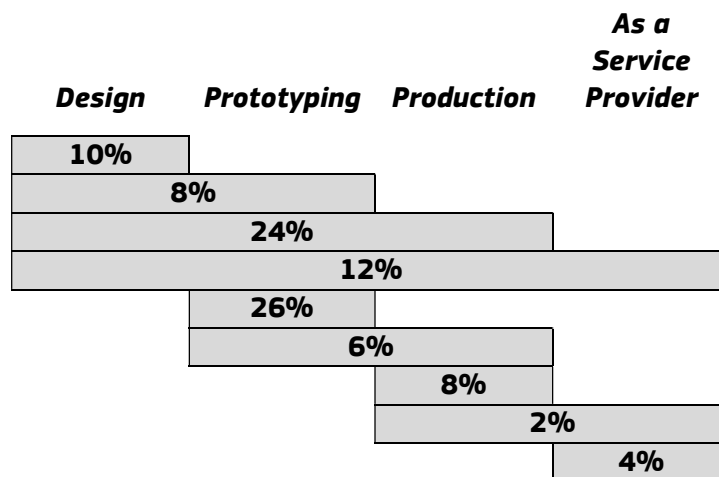
Several technology providers are involved in service design as a secondary sector. This is because the introduction of technologies often involves the identification and organization of actors, methodologies and implementation practices. The opposite is not true, considering that the design of services identifies possible technologies suitable for the implementation of projects, but does not deal with development, prototyping and testing.

The marketplace is much more present as a secondary activity. The main reasons given relate to the ever-increasing specialization of products / services, resulting in little interest in proposing standardized solutions that can be marketed according to traditional schemes and flows.

43.3% of respondents said they are engaged in B2B services, while only 21.7% are engaged in B2C services. The hybrid approach which covers the remaining 35% is very interesting.

The main share of personalized services is covered by Bespoke Design, followed by a hybrid Bespoke / Custom Design approach applied by businesses with online personalization services.

The specialization field reports some very interesting data which are summarized below. Of the 83 companies surveyed, 37 declare that they operate in material innovation, 18 in sustainable materials and 14 in digital services. There is a strong interpenetration of skills for those who deal with advanced materials, as it has been repeatedly stated that this specialization is flanked or intersects with additive manufacturing, mainly, as well as with smart textiles and wearable technologies. 13 companies are engaged in production and services for Accessible Design / Inclusive Design / Ethical Design / Design-for-All, but only 4 operate exclusively in this field.



Overall, fibres and advanced textile materials are the most used materials, 22 out of 83 companies declare that they are engaged in their development and production. Composite materials follow. Of little use and application of materials such as gel and foams, light alloys and coatings, probably motivated by the fact that this specialization is often reserved for larger companies and that application in creative companies is not immediately usable.

39 companies declare that they are users of digital technologies in their production and services; of great interest is the data relating to Digital Customer Engagement, 26 subjects declare to involve customers within the production process, with the possibility of customization, choice or booking.

The prevalent use of additive manufacturing technologies is in the prototyping stages, with 39 companies. Following in the design services with 31 companies. 27 respondents declare that they also use these technologies for finished products, but only 4 of them do it exclusively, that is, they are companies engaged in the development and direct sale of goods made in 3D Printing. Few are the realities that are configured as service providers. In terms of the most widely used additive technologies, there is a general equitable distribution of all those surveyed, with the prevalence of Fused Deposition Modelling technologies.

Finally, it is noted that 43% of the interviewees operate as Original Design Manufacturer. Some of them operate as Original Brand Manufacturers, 8.4% exclusively. The same percentage is found for Original Equipment Manufacturers, demonstrating that only a small part of creative companies operate in this field.

4.4.2 Emerging Hubs

From the census carried out within the database and from the data provided by the companies in the survey, some territorial specializations clearly emerge, often linked to urban agglomerations as well as to territorial areas spread over territories with a regional or national extension.

4.4.2.1 Sustainable and Tech Fashion in Amsterdam

Over 100 fashion houses are actually situated in Amsterdam and the city is also home to various major fashion labels' European headquarters. The fashion industry has a long way to go before becoming sustainable, but the awareness of sustainability issues is widespread in the Netherlands. Dutch consumers are generally very conscious. In the last years Amsterdam is building a growing reputation of being at the forefront of technology and sustainability, with startups concentrating on 3D digital design, new manufacturing techniques, and biodesigned textiles, there is a lot of innovation and startups working towards a circular approach. The main elements that corroborate these considerations derive from the number of companies and sustainable trajectories, with particular reference to:

- Textile recycling. Some clothing collection companies take responsibility for fabrics donated to clothing banks. The goal is that each new item of clothing produced contains a percentage of used fibers.
- Textile innovation. Several startups work on material development, functionality and aesthetics and have their own knitting and innovation lab, atelier, specialists and textile experts to achieve this. Part of the solution for a sustainable fashion industry is finding ways to create exciting fabrics in new ways, and using the circular economy to create a design system that uses waste products rather than virgin ones;
- Marketplace for end-of-seasons fashion. Company that work with the fashion industry to ensure that every item of clothing is worn, helping to find a buyer for every piece already produced, and providing the tools brands need to develop lines that they can have confidence

will sell. When selling end-of-season collections, fashion houses retain full control over pricing, merchandising, and visibility of their excess inventory.

Amsterdam also hosts Fashion for Good, the world's first museum dedicated to sustainable style innovation, including shops and spaces that act as an accelerator for fashion startups wishing to grow.

Innovative firms in this ecosystem include Wieland, By Borre, Otrium, Noumenon, New Industrial Order; Lena the fashion library; Anouk Wipprecht; Iris Van Herpen; Minois; Osier.

4.4.2.2 Berlin as capital of wearable technologies

The case of Berlin is certainly an example of great interest in terms of functional aggregation of companies that invest in new technologies and related applications for creative companies, with particular reference to the fashion and home living sector. According to the data reported on the Berlin Fashion Week page, more than 50 companies in Berlin are already developing applications for smart wearables. By organizing competitions and offering prize money (Smart wearables competition / CreativeTechnology competition in 2015) and an accelerator program on smart wearables and smart textiles, the Senate Department for Economics, Energy and Enterprises, contributes to increasing the potential of smart wearables applications in Berlin.

This concentration of knowledge and young entrepreneurial realities has allowed the organization of conferences and exhibitions such as Wearable Europe, focused on wearable technologies, applications, and their commercialization progress. This event brings together material suppliers, component providers, manufacturers and end users. The main specializations of interest for our study are related to:

E-textiles: Products, Markets, Materials and Manufacturing, with application in fashion industry specialized in sportswear, health & fitness.

Electronic Skin Patches: The change in sensor form factor has opened vast new opportunities for creative industries, fashion firms are working on innovations enabling new easy-to-wear designs.

VR, MR and AR: technological development in displays, optics, microprocessors, sensors and connectivity at service of creative industries and markets to improve or interact in a new way with everyday products.

Innovative firms in this ecosystem include Moon Berlin, Kobakant, ElektroCouture, Trafopop, VOJD Studios, Motionlab.Berlin, Wear It, Wint Design Lab, Design Farm Berlin.

4.4.2.3 Interview to Anastasia Zagorni – Design Farm Berlin

Box 10 – Anastasia Zagorni

Anastasia Zagorni is an Innovation Manager with more than ten years of working experience in technologies such as material science, organic electronics, coating processes and design. She is currently active as a Head of the Accelerator DesignFarmBerlin – Design-in-tech Accelerator. She conceives and implements the program for innovative design start-ups and works together with founders on the development of business models, partnerships and financing strategies. Anastasia Zagorni stands for the view that technology cannot be taken for granted and only becomes tangible in the discourse of design and accessibility.

1) Berlin seems to be an important European hub for wearable technologies and electronics applied to clothing, textile and fashion retail. Is it a result of a structured programme of the last years (top-down) or come from an informal concentration of companies and experts?

It's primarily due to concentration of companies, experts, design schools, experimental creative groups. Also, it's due to the connection to other European hubs and the initiatives on the European level like STARTS.

2) Who is leading? Is there a coordinator? Do you think that "Design Farm Berlin" is one of the main actor in this way?

There is no coordinating group. There are several players from research, design & business domains. DesignFarmBerlin has been supporting several undertakings which evolved into companies or research groups e.g. MotorSkins, UXFTT (Esther Zahn), Same (Aeneas Stankowski), Smed.Wagner, Panorama Fabrics, Wint Design Lab.

3) Have you a list/web page with some numbers of this phenomenon? How many SME's and startups work in this sector?

For statistics you can get in touch with Berlin Partner of Economics and Technology GmbH. This is a unit for economic development. They accumulate data for various sectors.

4) "Design Farm Berlin" is the only tech accelerator?

There are several accelerators around which work with technology. Our focus has been on "design in tech" which means we supported designers who use technologies in a sensible way for people. This is when innovation happens. Technology alone is just a tool.

5) What about public funding or private capital flows? Do you have systematic access to external funding? Are the SME's helped in the first years?

We are a part of the Weißensee School of Art and Design. We are funded through public grants.

6) What's the level of activities? R&D, Innovation, business creation (acceleration / incubation), investment, training / academia, communication / positioning, commercial? All the activities are covered by central organization?

As a part of the Art school we cover the field of entrepreneurship/innovation. We are now gradually extending our activities toward research.

7) Which are in your opinion the most promising realities for the future? Can you suggest one/two case studies?

There are various case studies: 1. We are currently developing an entity, a cooperative, that would closely work with the school but will also have independent scope of activities toward research & innovation 2. Co-working spaces like Motion lab that host companies for a certain period of time and provide access to machines. 3. Wint Design Lab - experimental small companies that explore technologies and processes.

8) Which are the funding flows? Does the tech accelerator invest in the projects? Are these projects sustainable?

We don't invest as a public organisation. We are collaborating with several groups on this. The projects are sustainable in the sense that people are either founding a company or are employed in the field they have been focusing on.

9) What about the business models? Can you suggest some innovative way to explode the results of the collaboration? Or is it based on traditional supply and selling strategies?

There is a strong urge in the creative community in Berlin to explore collaborative models, other ways of organising.

4.4.2.4 The Finnish Additive Manufacturing Ecosystem

The Finnish Additive Manufacturing Ecosystem (FAME) is an innovative industrial ecosystem that aims to increase the role of 3D printing and bring Finnish players together in the field of additive manufacturing. Finland has positioned itself as a highly educated, high-tech country with stable infrastructure, competing in traditional labor-intensive manufacturing industry. These qualities have supported initiatives to position the country as a forerunner in the field of AM. By its nature, AM is a high value added production method, which is based on a highly skilled workforce and a high level of automation. In the future, artificial intelligence will also play an important role in both design and production, complementing the existing Finnish experience. Artificial intelligence applications already exist that support quality assurance in the AM process and Finnish research facilities are contributing to this development. All these attributes support the idea that Finland is one of the main AM hubs in Europe. The seemingly remote geographic location is not an obstacle due to the high value per kg ratio in the AM parts, unlike traditional low value added manufacturing where the cost of logistics is significant.

The goal of FAME is to increase the use of additive manufacturing technologies in the Finnish machine manufacturing industry and, consequently, to fuel innovation scenarios in highly strategic sectors such as product design, starting with furniture. The actors of the FAME ecosystem could, in the coming years, contribute significantly to the European market of products designed and manufactured in AM and of production systems, also thanks to elements such as low barriers to collaboration. This ecosystem will be based on the development and acquisition of new skills, aimed at creating a portfolio of systemic offers from which the world of creative companies could draw significantly, increase their AM use rate and share experiences, information, data, physical equipment and facilities.

Innovative firms in this ecosystem include Kalevala; Hernesaari; UPM; Prenta; Materflow.

4.4.2.5 Interview to Markus Korpela - Finnish Additive Manufacturing Ecosystem

Box 11 – Markus Korpela

FAME M.Sc. (Eng), works at DIMECC Ltd. as FAME (Finnish Additive Manufacturing Ecosystem) Ecosystem Lead. FAME's purpose is to increase the role of additive manufacturing (AM) and bring together Finnish players in the field. Markus developed an interest in AM in 2015 and completed both his mechanical engineering theses about the topic, M.Sc thesis being about material needs of the Finnish manufacturing industry from the perspective of AM. Markus has experience from manufacturing industry at Rolls-Royce (curr. Kongsberg), and from academia at LUT University, where he also started his AM-related doctoral studies, at research group of Laser Materials Processing and Additive Manufacturing. His studies continue at University of Turku.

- 1) FAME seems to be an important ecosystem to join technology providers in 3D Printing and industrial players. Is it a result of a structured programme of the last years (top-down) or come from an informal concentration of companies and experts?

FAME was established in Autumn 2020. It is partly a structured programme initiated by 20+ companies and their AM (additive manufacturing, 3d printing in colloquial language) Experts, but it is also supported by the Finnish ministry of employment and economic affairs in the spirit of Public Private Partnership where public funding is provided for them, who first make a private investment.

- 2) Who is leading? Is there a coordinator? Do you think that "FAME" is the main actor in this way? Have you some connections with other European countries?

FAME is facilitated by DIMECC Ltd. FAME will co-operate with European AM ecosystems, and we have good connections to leading AM centre in EU. The collaboration has already taken the first steps with Italian, German and Swedish actors.

- 3) Have you a list/web page with some numbers of this phenomenon? How many SME's and startups work in 3D Printing sector?

In Finland, we have some tens of SMEs in the sector.

- 4) Have you some numbers about the role of "FAME" for creative sector? How many creative industries work with "FAME"? Have you some interesting case studies about furniture, lighting, fashion products realized in collaboration with your network?

FAME doesn't, but there several companies who make business with AM in furniture, art, and jewellery sectors in Finland.

- 5) What about public funding or private capital flows? Do you have systematic access to external funding? Are the SME's helped in the first years?

FAME is funded by current ecosystem members and granted public funds

- 6) What's the level of activities? R&D, Innovation, business creation (acceleration / incubation), investment, training / academia, communication / positioning, commercial? All the activities are covered by central organization?

R&D, co-creation, co-innovation, training, benchmarking. Activities are performed by the ecosystem members, central organisation DIMECC only facilitates the activities.

- 7) Which are in you opinion the most promising realities for the future in 3D Printing? Can you suggest one/two case studies?

Automation in AM production -> serial production, new materials, tailored microstructures, medical sector applications

- 8) Which are the funding flows? Does FAME invest in the projects? Are these projects sustainable?

FAME acts only as an ecosystem. All the projects are decided and invested case by case by the members.

- 9) What about the business models? Can you suggest some innovative way to explode the results of the collaboration? Or is it based on traditional supply and selling strategies?

FAME bases on knowledge sharing and RDI projects executed together. That way we bring value to our consortium members.

4.4.2.6 Portugal Footwear Cluster

The footwear industry is particularly relevant in the north of the Portugal, where the majority of the companies are based. The industry has two major production centers: one in Felgueiras and Guimarães; and the other one in Santa Maria da Feira, São João da Madeira and Oliveira de Azeméis. Both hubs are approximately at 50/60 kilometers from Porto. A third, smaller, center is located at Benedita, at 90 kilometers from Lisbon. Some facts give us an idea of the size and importance of the two larger footwear clusters for Portugal in general, and the northern region, in particular: the footwear sector accounts for 6,7% of the jobs and 2,8% of the business volume in Portugal. The figures increase to 12% and 6.8%, respectively for the northern region. The footwear industry is, therefore, one of the main economic foundations of the northern Portugal.

The geographical concentration of companies facilitates the relations between firms specializing in the manufacture of footwear, components and leather goods, creating good conditions for formal and informal business networks, and it enables firms to benefit from economies of agglomeration. In addition to the business fabric, some of the sector's main infrastructures, such as the Portuguese Footwear Technology Center¹⁴ and the Design and Footwear Academy (CFPIC),¹⁵ are present in both centers mentioned above.

According to the 2019 Statistical Report by the Portuguese Footwear, Components, Leather Goods Manufacturers' Association APICCAPS,¹⁶ the Portuguese footwear cluster is formed by 1877 companies, of which 48.3% of which are micro-enterprises and 39.5% are small. Only 11.7% of firms employ between 50 and 249 workers (medium-sized businesses) and 0.6% more than 250 (nine footwear manufacturing companies, one in components and one in leather goods). The foregoing data leads to the conclusion that the average size of firms is very different in the three industries that make up the cluster. A footwear manufacturing company does in fact employ an average of 26.4 workers, a number that decreases to 20.5 in the components industry and 15.6 in the leather goods industry.

The sector seems to be characterized by a high capacity of flexibility and quick response, where even small enterprises demonstrate being able to respond to a very selective demand with a high quality of production. This achievement is possible thanks to the adoption of advanced technological solutions for their production processes. In terms of innovation

¹⁴ Centro Tecnológico do Calçado de Portugal <https://www.ctcp.pt/>

¹⁵ Centro de Formação Profissional da Indústria do Calçado <http://www.cfpic.pt/index.php>

¹⁶ Associação Portuguesa Ind. Calçado Componentes Artigos Pele Sucedaneos (APICCAPS) <https://www.apiccaps.pt/>

provider, the Portuguese Footwear Technological Center (CTCP) plays a key role developing and disseminating a constant flow innovations. CTCP has taken on the responsibility of promoting partnerships between footwear companies, technology suppliers and institutions from the technological and scientific system to develop ambitious innovation projects. This network approached has allowed the Portuguese footwear industry to exploit an innovation potential which is unusual for a sector mainly composed by small and medium-sized enterprises. One successful example of a Portuguese technology was the development and introduction of cutting leather with a water jet, which allowed for high production flexibility without compromising on price.

It is also worth mentioning the approach of young creative companies to change design methods. According to the EU “Digital Fablab” project,¹⁷ around 35% of workers are over 50 years old, making it essential to attract motivated and qualified young talent with new and innovative VET (Vocational and Educational Training) practices, reducing the skills gap between education and work, and adapting to an increasingly digital world. By simulating real-world experiences, in Portugal cluster are going to born fully digital FabLab using Augmented Reality to teach and learn practical shoe manufacturing skills.

Furthermore, the Portuguese cluster is very active in accelerating the interaction of the ecosystems along four different dimensions. On the design axis, by developing specialized solutions for designers; on modelling one by establishing a system's architecture by modules introducing advanced modelling of soles and customizable elements. Furthermore, on production side by introducing additive production systems and, last, on marketing promoting innovative solutions for commercializing the manufacturers' offer. At the same time some cluster projects investigate the way shoes can be presented to customers and consumers through a low-cost multimedia information system specially prepared for small businesses without 3D CAD systems.

The materials ‘front’ is oriented towards the recovery of waste material generated in the manufacturing process and towards new concepts of comfortable, economically competitive shoes made of leather and biodegradable / compostable components.

Finally, the cluster is active in integrated 3D scanning and shoe customization approaches, by using 3D foot scanning systems to create a custom mould to build the shoe upon. This process is designed to develop footwear that flawlessly follows each curve of feet to ensure fit and comfort. These systems are integrated with customization software for the selection of styles and colours, up to the creation and production of perfectly ergonomic and replicable footwear.

Innovative firms in this ecosystem include: Aloft, Pilar Shoes, Nae, Plano8, Zouri, Wayz, Skyzo, Shoevenir.

4.4.3 Original keywords

While collecting information on the companies, we collected pieces of information on novel application routes, ingenious approaches, inspiring outlooks, forward looking insights, etc. that may offer insights on upcoming trends. In the following we present a number of ‘original keywords’ provided by the analysed companies surveyed. The result assumes a certain relevance in terms of approach and services and evolutionary vision of objects:

¹⁷ <https://www.digitalfablab.eu>

- **From waste to value:** claim that does not only belong to creative companies, but that characterize website and communication of material providers/manufacturers engaged in the recovery and valorization of pre and post-consumer waste deriving from the manufacturing, agro-industrial and fishing industries;
- **Technospiritualism:** polydisciplinary activity based on working with unstable media, experimental interfaces, electronic textile and augmented materials. Technospiritualism investigates sensorial and conceptual relationships between the individual and its physical surroundings, aiming to create experiences that incite a personal process to the user. Creative companies tend to investigate liminal states, notions of time, and the state of presence, using technological methods as active, participatory installations or multi-sensorial performances;
- **Sensual Tech Startup:** companies that encourage exploration, experimentation and expression of sensual experience and sensual wellbeing, focusing on materials, technologies and new UX directions;
- **Look beyond the garbage:** companies that encourage to overcome the cultural barriers deriving from the use of recycled materials or materials destined for landfills. This approach has a reassuring value on the quality and the prospective importance of the intrinsic value of each object.
- **Poetic technology:** companies that are engaged in transformation of complex technologies into meaningful objects or scenarios where beauty and functionality goes hand in hand. They aim to create sustainable futures, not only by environmental point of view but in terms of sustainable integration and use of technologies, with reduction of invasiveness in everyday life;
- **Emotional accessories:** high-tech products, but particularly jewels and fashion accessories, with an attractive design and added value reflecting user emotions showing some changes, as for example colours. These kinds of companies are working in technologies able to change the characteristic of the accessories according to multiple factors, defined by the user or by the logic of the same products that analyses, interpret and then shows user's emotions and mood.
- **3D-Printed ready-to-wear** - Collections, often the result of 3D scans of natural elements and structures, which replicate complex three-dimensional geometries in 3D printed clothing, thanks also to the possible digital combinations of rigid and rubbery materials. These blends of textures create garments that emulate the movement of wearable fashion garments and are therefore ready to wear.
- **Conscious Luxury** – Companies and relative brands that want to combine the ethical, the timeless, and the beautiful in the luxury market. They usually care about people, animals and planet, products are made in Europe, using sustainable materials, where no animals are harmed in the design and manufacturing process. Their mission is to transform the fashion industry, with a focus on sustainability, transparency and quality.
- **Therapeutic Textile Products** – Market innovative therapeutic textile products that provide comfort and happiness to people with a variety of sensory needs, particularly those living with autism, dementia, and anxiety. New and emerging technologies have the potential to support care in the home and this is the area in which some companies are able, using Smart Textile technology, to provide therapeutic interactive solutions;

- **VR/AR Try-On Solutions** - Thanks to a mobile app, companies that propose Virtual Try On technology allows to try any product without physically having it. It is possible to virtually choose and configure eyewear, a pair of shoes, as well as a dress, a bag or any wearable accessory. Try On solutions prove to be a precious tool for the commercial strategies of Fashion and Luxury companies.

List of abbreviations and definitions

3D	Three Dimensions
AM	Additive Manufacturing
AR	Augmented Reality
B2B	Business to Business
B2C	Business to Consumer
CAD	Computer Aided Design
CCIs	Cultural and Creative Industries
CNC	Computer Numerical Control
DfAM	Design for Additive Manufacturing
FDM	Fused Deposition Modelling
ICT	Information and Communication Technologies
IoT	Internet of Things
MR	Mixed Reality
OBM	Original Brand Manufacturer
ODM	Original Design Manufacturer
OEM	Original Equipment Manufacturer
RIM	Reaction injection moulding
RTO	Research and Technology Organization
SLS	Selective Laser Sintering
SMEs	Small and Medium Enterprises
TCF	Textile Clothing Footwear
UI	User Interface
UX	User Experience
VET	Vocational and Educational Training
VFX	Visual Effects
VR	Virtual Reality

List of boxes

Box 1 - Startups using Additive Manufacturing technologies for furniture, lighting or interiors.....	12
Box 2 - Startups using Additive Manufacturing technologies for textiles, clothing, footwear, jewellery, fashion or accessories.....	14
Box 4 - Empty your Wardrobe	22
Box 5 - Castor Sunglasses	22
Box 6 – MycoTEX	23
Box 7 – Mobibam	26
Box 8 – Iris van Herpen / Ganit Goldstein	26
Box 9 - MareTech.....	29
Box 10 – Anastasia Zagorni.....	41
Box 11 – Markus Korpela	43

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